Multicubator

Team: 10209

Grade 11 semester 1 (2025/2026)

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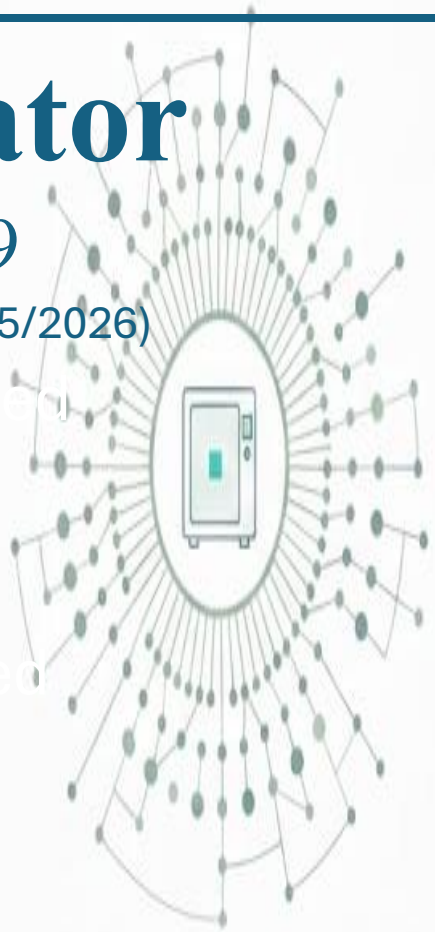


Table of Contents

Chapter 1 4

Egypt grand Challenges 5

Improve the scientific and technological environment for all 5

Address and reduce pollution fouling our air, water, and soil 9

Reduce and adapt to the effects of climate change. 13

Recycling garbage and waste for economic and environmental purposes 17

Problem to be solved 21

Increasing the capabilities of research in Egypt's laboratories through recycling electronic waste 21

Research 25

Topics related to the problem: 25

Topics related to the solution: 26

The prior solutions 27

OM118-M3 40X-400X Monocular Student Compound Microscope 27

VEVOR 20L lab incubator 32

Thermo Scientific Heratherm IGS180 36

Ohaus Guardian 2000 Stirrer | e-G21ST04R 41

Chapter 2 45

Solution requirements 46
Design requirements 47
The selection of the solution 48
Selection of Prototype 49
Chapter 3 51
The abstract 52
The introduction 52
The materials and methods 53
Test Plan: 56
Data collection 56
Chapter 4 59
Analysis and Discussion 60
Binary fission formula 60
Turns ratio equation 61
The Peltier law equation 62
The conclusion 64
Recommendations 64
1. Humidity control system recommendation 64
2. CO2 injection system: 65
3. Water jacket insulation system: 65
4. stainless steel & copper lining: 66
Recommendations for future teams 66
Project Benefits 67
The literature cited 67

Chapter 1

Egypt grand Challenges

Egypt faces many problems in various aspects, the Egyptian government does its best of to fix these problems. All these problems can be summarized in "Egypt Grand Challenges". Egypt's Grand Challenges are the most important problems that prevent Egypt from growth and development in economy, life quality, …. etc. The government Tries to fix Egypt's Grand Challenges by introducing initiatives, implementing innovative ideas to solve the problem at the lowest cost, …. etc. Some important Challenges **are shown in Figure 1.1**.

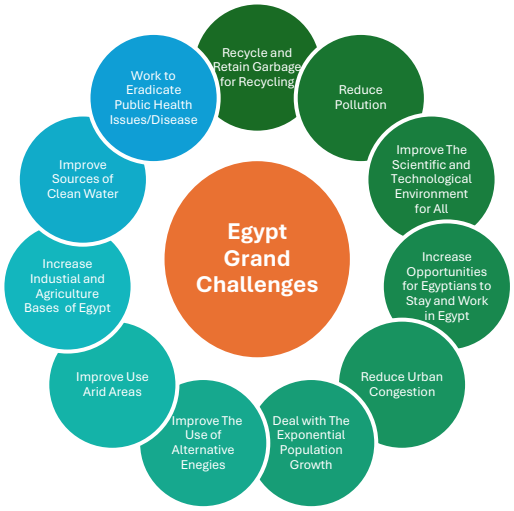


Figure (1.1): Egypt grand challenges

Improve the scientific and technological environment for all

Another important challenge for Egypt is the scientific and technological environment in Egypt. Science and technology are the keys to prosperity; Therefore, Egypt must improve the scientific and technological environment for all. Egypt will rank 79th in the Global Knowledge Index (GKI) in 2023, up from 83rd in 2020, the Ministry of Education said. Furthermore, Egypt ranks poorly in education, scoring 49 out of 50 countries for grade 4 students in reading. And 34 out of 39 countries for students in grade 8. In addition, its overall ranking in the Human Development Index (HDI) is 97th globally, reflecting challenges in educational outcomes. These statistics indicate Egypt's lack of education. Illiteracy in Egypt fell to 16.1%; However, it is still considered high compared to other countries. According to Statista, the number of technical and vocational education and training (TVET) graduates in Egypt in 2021/2022 was approximately 640,000. Of these, 285,500 have specialized in industrial technical education, as shown, 62.7 percent were

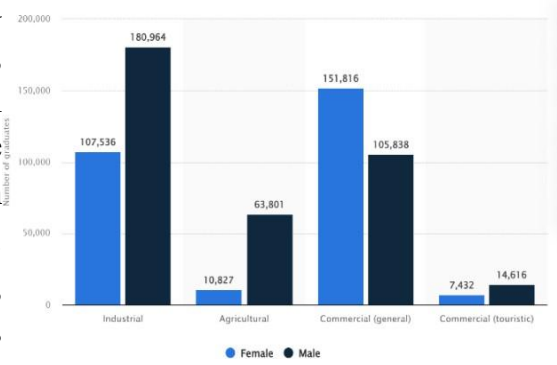


Figure 1: The number of TEVT graduates in the 2021/2022 year

men. **Figure 1**. In addition, Egypt contains 10 technical universities. compared to According to TU9, there are 17 technical universities in a country like Germany. This reflects Egypt's lack of technical education.

causes

1. Quality of education

Later, literacy increased in Egypt. However, as shown in **Figure 2**, it is still considered low compared to developed countries. There are a few reasons for the low price. Ancient Curriculum: The curriculum in Egypt focuses on memorization more than critical thinking and problem-solving skills. It also depends on old concepts that are not favorable today. Private tuition: According to the ERF, an average of 40% of students receive private tuition, and this percentage increases to 60% for upper secondary level students. 44% of students in urban areas take private tuition, while in rural areas the figure is 35%.

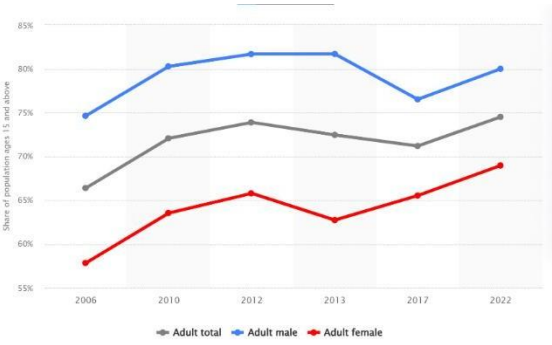


Figure 2: Increase in literacy in Egypt

2. Quality of public schools

According to CIRA, due to the high number and low quality of Egypt's public schools, as shown in **Figure 3**, private schools in Egypt account for approximately 10.6% of the total student population. The number of private school students in Egypt has increased by 6.3% over the past five years, far exceeding the number of public school students, which has increased by 3.6%.

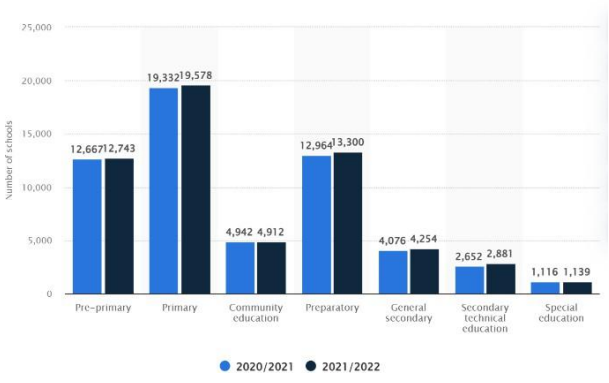


Figure 3: Number of schools in Egypt as of 2021/2022, by educational level

3. Technological and digital separation

In Egypt, the curriculum focuses on teaching older technology concepts, such as Windows 7, and older programming languages that are not as important these days. Furthermore, many students, especially in rural areas of Egypt, do not have access to technology outside of Egypt, which negatively affects the country's future. Some programs are trying to improve the knowledge of Egyptians, such as the Digital Egypt Cubs initiative, which enrolled approximately 7,090 students in March 2023, according to DECI, but this is still not enough to achieve a thriving society.

Impacts

1. High unemployment

According to technological advances worldwide, technical jobs such as programmers and cyber analysts are required. And according to the digital divide in Egypt, not many people work in these sectors.

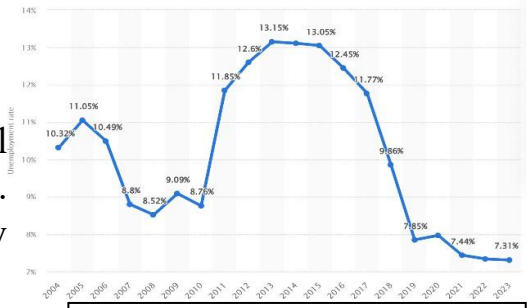


Figure 4: Unemployment rate from 2004 to 2023 in Egypt

As shown in **Figure 4**, the unemployment rate in Egypt is projected to reach 7.31% in 2023 according to Statista. What is higher than the global unemployment rate? As shown **in Figure 5**, the global unemployment rate has reached 4.96% according to Statist

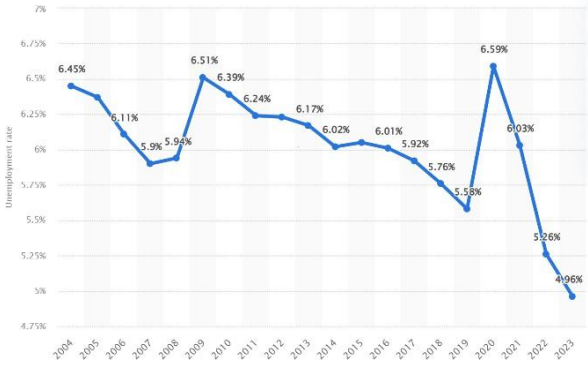


Figure 5: Global unemployment rate from 2004 to 2023

2. National security risk

Egypt has weak cyber defense systems, leaving banks, government, and infrastructure vulnerable to cyber-attacks if there are no cybersecurity personnel in place to protect them. According to Statista, by 2024, the internet penetration rate in Egypt is expected to reach 72.2 percent, up slightly from 37 percent in 2017, as shown in **Figure 6**.

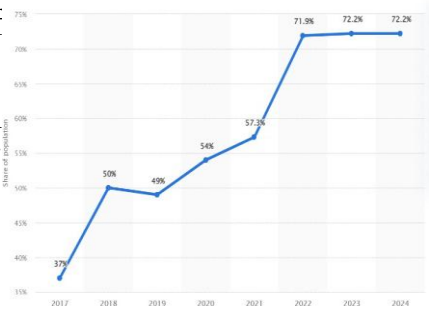


Figure 6: Internet user penetration in Egypt from 2017 to 2024

In addition, cybersecurity threats are also increasing as 90% of Egyptian companies face cyber risks (Egyptian Ministry of Communications, 2022).

3. Cultural and knowledge influence

According to UNESCO in 2023, only 30% of Egyptian students have access to digital learning tools and websites. In addition, research results in Egypt are low; Therefore, Egypt ranks 86th among 132 economies in the Global Innovation Index 2023 according to WIP. Furthermore, according to the Academy of Research and Technology, Egypt has reached 55th place in the global research and development (R&D) sub-index, according to the 2023 Global Innovation Index report. Furthermore, Egypt's annual GDP growth is shown in **Figure 7**.

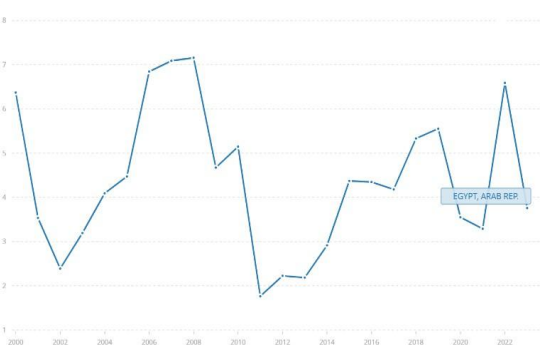


Figure 7: GDP annual growth in Egypt

Address and reduce pollution fouling our air, water, and soil

Pollution has been a danger in the last decades, threatening the fate of humans all around the globe. Egypt, being one of the most crowded countries in Africa, it is one of the most countries affected by pollution. Doesn't it affect only people, but pollution also destroys crops, kills marine creatures, and destroys whole ecosystems. Reported by the ECESR, unsafe drinking water and poor sanitation contribute significantly to kidney and liver diseases, schistosomiasis, and diarrhea. WHO estimated 5.1% of deaths and 6.5% of disabilities to be caused by those factors.

Cairo stands as the most pollution-afflicted governorate in Egypt, grappling with severe levels of air, water, and solid waste contamination due to its high population density, concentrated industrial activity, and inadequate waste management infrastructure. The annual nationwide average PM2.5 concentration in 2019 was 13.6 times greater than WHO 2021 recommended concentration levels. As demonstrated in **Figure 1,** road transport is the most prominent cause of PM in Egypt. This makes densely populated cities more vulnerable to pollution, placing Cairo at the top of the list of Egypt's most polluted cities, followed by Giza and Qalyubia.

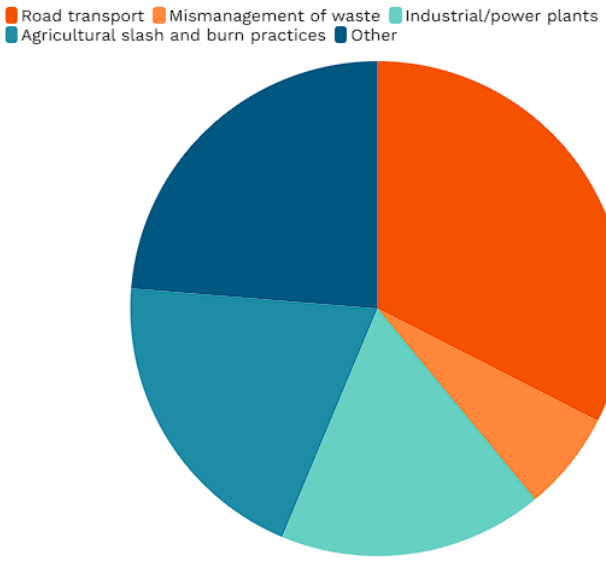


Figure 8: Major factors of PM in Egypt

Causes:

1. Vehicle emissions

Road emissions have been the main factor of pollution in Egypt, causing severe damage to both individuals and the government. The primary source of PM2.5 air pollution in the greater Cairo area is road transport, accounting for a devastating 33% of the total. This is caused by aging private cars and taxis, which account for 80% of the extreme congestion problem. Traffic volumes can reach 7,000 vehicles/hour per lane, far exceeding the maximum theoretical flow of 1,900. As shown in **Figure 2**, emissions from transport (primarily cars) continued to increase from 21.2 MtCO2 in 2005 to nearly 27 MtCO2 in 2012, showing the severe damage cars and transport can cause to our air.

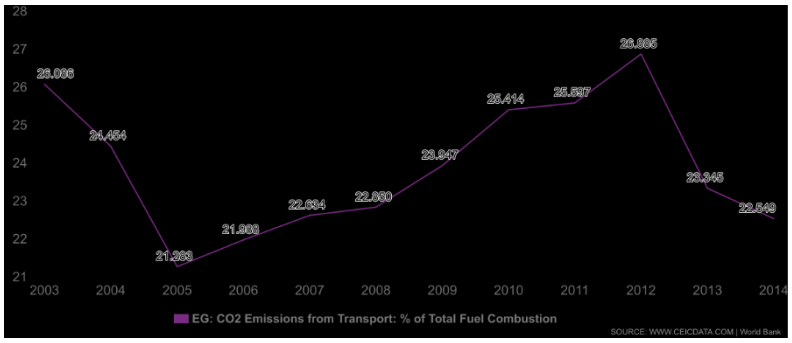


Figure 9: Egypt road transport emissions 2003-2014

2. Industrial effluents and heavy metals

Industry is one of the pillars of our modern life, but it is also a major cause of pollution. Reported by the water research and pollution control department in Cairo, about 350 factories dump wastewater directly into the Nile or through drains; Alexandria alone contributes around 60% of the coastal marine pollution via 1250 plants discharging into Lake Mariout. As represented in **Figure 3**, EEAA and current studies report that large volumes of plastics and heavy metals persistent organic pollutants (PCBs) have been tracked across the Nile,

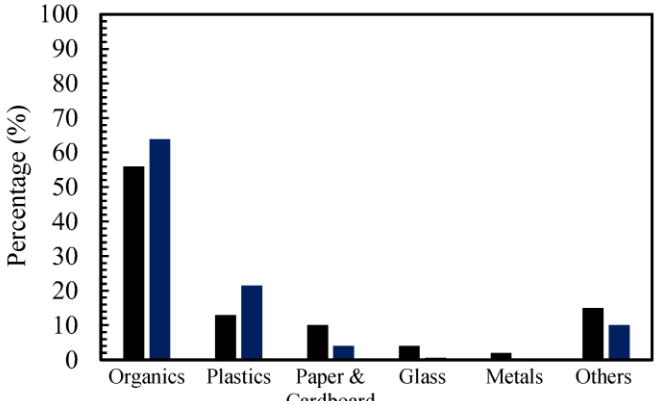


Figure 10: Comparison between the waste composition in Cairo (current study) and the generalized composition in Egypt issued by the Egyptian environmental affairs agency

meaning that they're a serious cause of water pollution for Egypt, especially the Nile.

3. Atmospheric Deposition & Dust

According to Assiut University, industrial dust from mining and factories carried by wind deposits heavy metals like Cd, Cu, Pb, and Zn on soils. As shown in **Figure 4**, Zn equivalent concentration already surpasses the critical level in 3 of 5 sites. According to the Egyptian Journal of Chemistry, a survey in Sinai and Ismailia found 20 pesticide residues, with total soil concentrations of 0.505 mg/kg in South Sinai, 0.075 mg/kg in North Sinai, and 0.207 mg/kg in Ismailia.

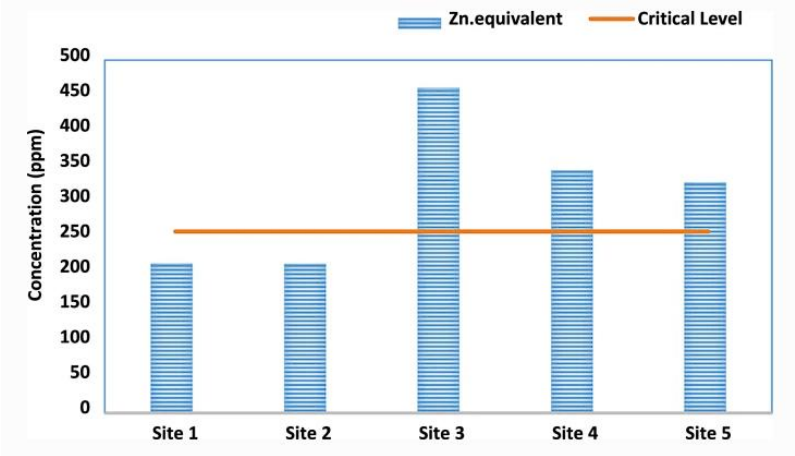


Figure 11: Zn equivalent measured in 5 different sites

Impacts:

1. Health Impacts – Leading Cause of Premature Deaths

Air pollution has many severe and widely recognized impacts. The most recognized one of all is its impact on public health. As shown in **Figure 5**, all the measurements of $PM_{2.5}$ exceed even the WHO annual guideline of 10 µg/m³, highlighting persistent overexposure for Cairo residents, resulting in chronic health burdens like respiratory and cardiovascular diseases. According to the WHO, in

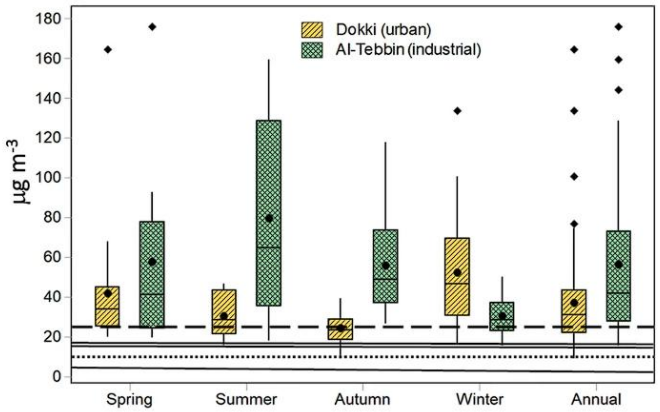


Figure 12: Box-and-whisker plots show higher $PM_{2.5}$ concentrations in industrial zones (Al-Tebbin) versus urban residential areas (Dokki),

2023, 49,756 deaths in Egypt were due to ambient particulate matter (PM) pollution; an additional 3,452 deaths from household air pollution, totaling around 53,200 deaths.

2. Human Health Risks Beyond Fish Consumption

In the Nile Delta, shallow aquifers are contaminated by nitrates, pesticides, and heavy metals. Residents using irrigation canals for drinking water have increased rates of **kidney, liver diseases**, and even **pancreatic cancer**, especially in northeastern Delta villages. Water pollution also contributes to waterborne and microbial diseases, as El-Batts drains and Fayoum irrigation canals show high fecal coliform levels (up to hundreds of thousands CFU/100 mL), especially in summer, contributing to **intestinal helminth infections** and diarrheal illness.

3. Contamination Hotspots & Ecological Risk

In **El-Minia** (near El-Moheet drainage), soil is contaminated by Zn > Cr > Cu > Pb > Co > Cd, with **70.7% of the area showing "considerable" contamination**, Cd being the most severe. **Fayoum's Eastern area** shows **very high ecological risk** from Cd (risk index ~4,157), moderate risk from Pb (~51), and lower risks from Ni and Mn. While adults have a non-carcinogenic hazard index (HI) below safety (<1), **children exceed it (HI >1),** with **carcinogenic risks from Cd, Ni, Pb above acceptable levels (>1×10^{-4}).**

Reduce and adapt to the effects of climate change.

Egypt's arid climate will suffer from additional environmental stresses because of climate change, including extreme temperatures, irregular precipitation, elevated sea levels, land subsidence, coastal flooding, shoreline erosion, deteriorating soil salinity, and persistent drought. These mutually reinforcing impacts will build on one another, worsening water scarcity, hindering food security **as shown in Figure 6**, displacing exposed populations, and destabilizing the Egyptian economy. Climate change will therefore exert added pressures on already precarious populations and push still other groups to unprecedented levels of vulnerability. Efforts to adapt to a changing climate will also exact a steep price, exposing existing governance shortfalls. Hot, dry, and consisting almost entirely of uninhabited desert, Egypt has long had environmental challenges. The North African country overlooks the Mediterranean and Red Seas, and almost all its freshwater comes from the Nile River. Owing to the arid climate, 95 percent of Egyptians live on less than 6 percent of the land, mostly around the Nile. 1 This population distribution has come about in part because Egypt's scarce annual precipitation is limited to a mean of 33.3 mm, with most centered along the Mediterranean coastline in Lower Egypt. 2 Rainfall in the country amounts to less than 1.8 cubic kilometers annually, of which only 1 cubic kilometer is successfully exploited for agriculture. It comes as no surprise that Egypt has

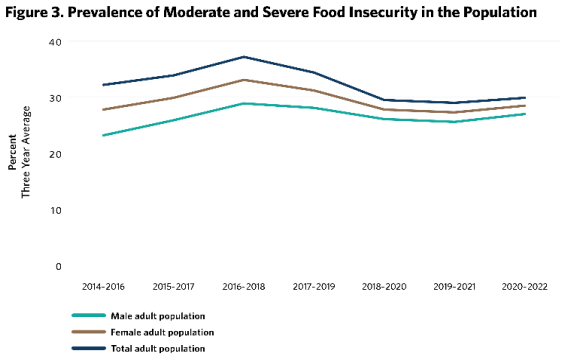


Figure (13): food insecurity through climate change

historically turned to, and settled around, the Nile for thousands of years. The current water situation is not sustainable. Once the breadbasket of the Roman Empire, Egypt today can no longer depend on the Nile to meet all its freshwater needs, particularly for irrigation. This shortfall is attributable in part to demographics and their consequences on water demand. Between 1989 and 2018, Egypt's population grew at a rapid rate of 2.1 percent per year, almost doubling in three decades to reach 109 million people in 2021.

Causes:

GHGs emissions:

Egypt's total GHGs emissions rose to 134 million tons (Mt) of carbon dioxide equivalent (CO_2) in 1990 to 342 million tons(Mt) CO_2 in 2030, as shown in Figure 7, representing a 0.73 % share of global emissions because of the factory smokes that increases gradually in Egypt, and another scientific factor for increasing the GHG gases is the CO_2 emissions per capita in Egypt that is equivalent to 2.40 tons per person (based on a population of 112,618,250 in 2022), an increase of 0.11 over the figure of 2.25 CO_2 tons per person registered in 2021; this represents a change of 5.0% in CO_2 emissions per capita.

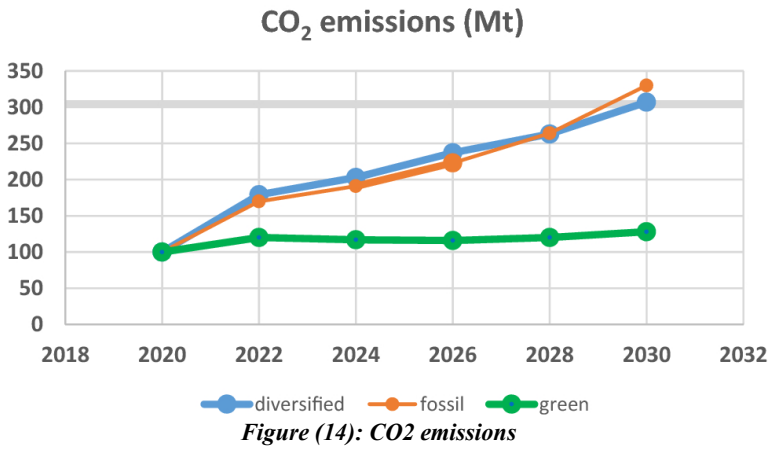


Figure (14): CO2 emissions

The production of energy in Egypt:

In Egypt, the electricity is produced from non-renewable energy resources, and about 85% of our energy is non-renewable, as shown in Figure 8. In 2022, Egypt produced between 600,000 to 800,000 barrels per day and made about 70 to 75 billion cubic meters of natural gas. Coal usage in Egypt is minimal compared to oil and natural gas, which release harmful greenhouse gases into the atmosphere, creating the greenhouse effect, which causes global warming. Non-renewable energy sources are also harmful pollutants and lead to habitat destruction, which produces carbon dioxide and nitrous oxide that trap Earth in the heat of the sun.

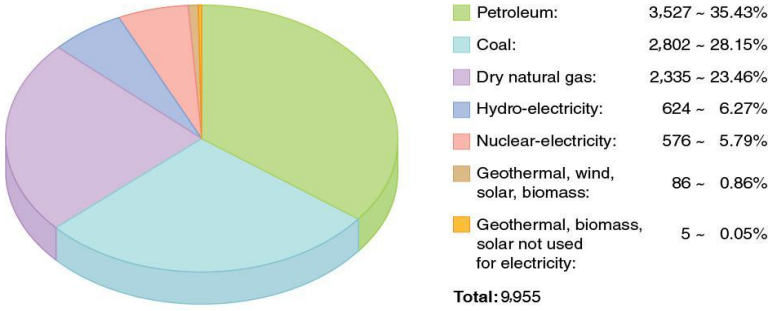


Fig 15: Energy production analysis in Egypt

Waste Mismanagement:

Increasing the use of plastic bags and using paper bags instead will significantly contribute to water pollution, as burning these materials increases the percentage of CO_2 and methane in the atmosphere. Waste mismanagement is a widespread issue that significantly impacts the environment, causing daily damage. Damage to the environment causes damage to every part of the environment: the air, water, and other necessary resources, and it will increase the temperature of the Earth. Pollution is one of the most significant disadvantages of waste management, as it helps maximize environmental contamination caused by waste. When factories began to use raw and fresh materials, the old materials were unused and went through landfills and oceans, and their weight reached 90 to 100 million very massive tons.

Impacts:

1.Water sacrifices:

In Egypt, water conservation has become an important part of the country after changes in the climate and temperature, and perhaps of greatest concern is the potential for a significant reduction in the flow of the Nile River. Also of major concern are the sea level rise and higher temperatures. Reduced water supplies, loss of land, and higher temperatures could adversely impact the economy, human health, and ecosystems in Egypt, as shown in Figure 9. The country is grappling with a severe water deficit, with an annual shortage of around 7 billion cubic meters, and the country could run out of water by 2025, when it is estimated that 1.8 billion people worldwide will live in absolute water scarcity (source: IPS), and climate change is a key part of the problem.

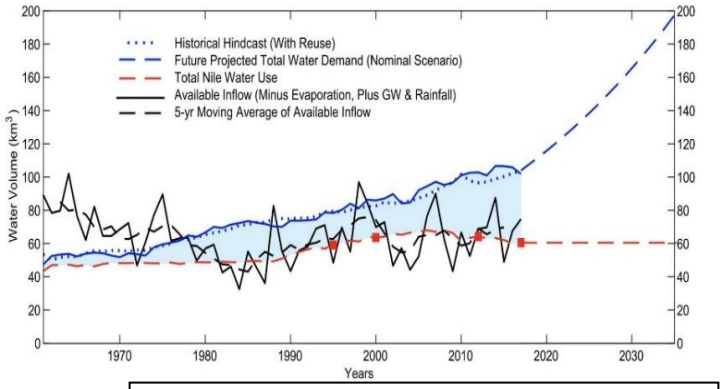


Figure 16: Impacts of climate change on water

2.Agricultural impacts:

Agriculture is the backbone of the Egyptian economy and food security, but it is very unsafe due to climate change. Rising temperatures and water shortages reduce the productivity of larger crops such as wheat, rice, and corn. Heat stress can damage crops. Saltwater infiltration. Growth on the sea and low Nile flow cause saltwater due to the infiltration of a large

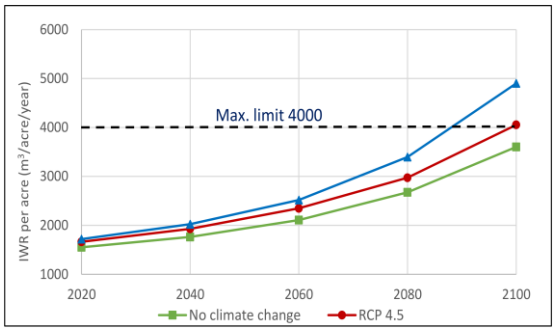


Figure (17): Effect of climate change on agriculture

agricultural area, the Indigo Delta. This saltwater of soil and water reduces the field's fertility. Deserts and urbanization reduce the number of countries that have access to agriculture. By 2100, up to 15% of the agricultural area of the Indigo Delta may be lost to growth in the sea level, as shown in Figure 10.

3.Natural factors affecting Egypt:

Egypt experiences more frequent and intense extreme weather events due to climate change. The first weather event is heat waves. Long and more intense heat waves become commonplace, affecting the demand for health, agriculture, and energy. The second is flash floods. Suddenly, heavy rainfall in dry regions can cause floods, damage to infrastructure, and the displacement of societies. The last thing is dust and sandstorms. Increasing temperature and desertification will increase the frequency of dust and sandstorms, affecting air quality and visibility.

Recycling garbage and waste for economic and environmental purposes

Egypt faces significant challenges in managing garbage, waste, and recycling, which have large environmental and social consequences. The country produces millions of tons of waste annually, with a substantial percentage being organic or recyclable materials. Egypt produces approximately **90 million tons of waste annually,** with **20 million tons** attributed to municipal solid waste (MSW). A smignificant proportion of this waste, **45% organic,**

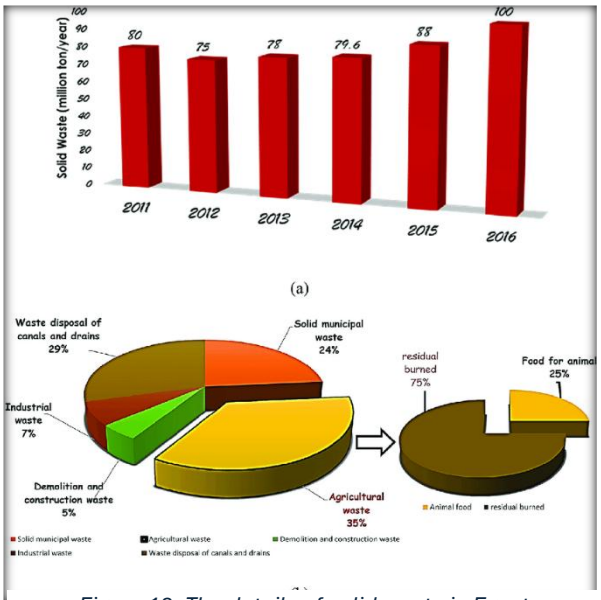


Figure 18: The details of solid waste in Egypt

14% plastic, and **4% glass and metals**, is recyclable. However**,** only **10-15% of recyclable waste** is processed due to limited recycling infrastructure. Egypt occupies the 7th position in the list of countries with the most mismanaged plastic waste, according to a recent report published in the *science* magazine in 2010.

The consequences of these challenges are in a lot of fields. Environmentally Insufficient recycling also leads to severe air and water pollution, particularly in urban areas, and impacts the Nile River ecosystem. Socially, the informal waste sector employs over **300,000 people**, but the lack of regulation results in hazardous working conditions and inefficient material recovery. The informal recycling sector, while critical, operates under unsafe and exploitative conditions, leaving workers vulnerable to health hazards. The overall loss from unrecovered recyclable materials is estimated at **$5 billion annually**. Additionally, limited public awareness about proper waste disposal further hampers efforts to improve recycling rates

To combat these challenges, Egypt is working on improving its recycling infrastructure, with plans to increase recycling capacity by **20% over the next decade** and treat at least **30% of organic waste** using biodegradable methods. The government is also introducing extended producer responsibility (EPR) schemes and aiming to cut plastic waste leakage by **20% by 2030**. Public awareness campaigns are essential components of this strategy, as only about **50% of the population** actively participates in waste segregation and recycling.

Causes:

1.Increased population growth rate

Egypt's huge population growth rate has been one of the major causes of the spread of waste and non-recycling. Although Egypt's population growth rate has declined in recent years by about 0.8%, as shown in **Figure 12**, it is considered a significant population growth rate. Egypt also occupies the 3rd position in Africa with an estimated population of 110 million, which increases the chances of keeping waste unrecycled due to the significant increase in waste amounts, making recycling waste a difficult process for the Egyptian government to fulfill.

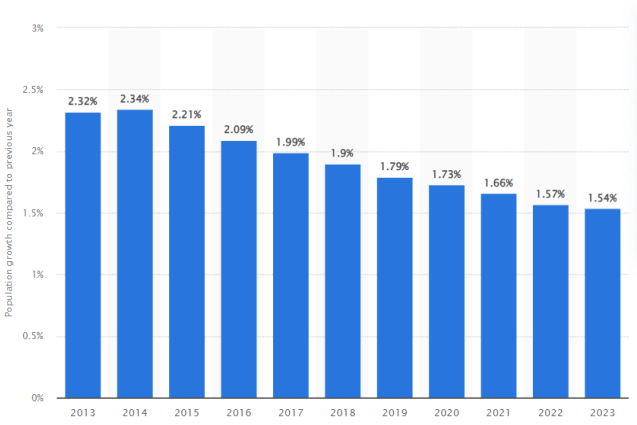


Figure 19: population growth compared to past years

2. Lack of public

Public awareness has declined in recent years in Egypt, leading to people throwing away huge volumes of waste in the dedicated places to be recycled. As shown in **Figure 13**, the recycling rate has been decreasing since 2014, because most people don't even know what can be recycled and what cannot, or even the benefits of recycling. This lack of knowledge leads to a shortage of recycling programs. According to AUC Knowledge Fountain, the recycling rate in Egypt does not exceed 20%, while in Europe and North America, the recycling rate is between 40-50%. In Egypt,

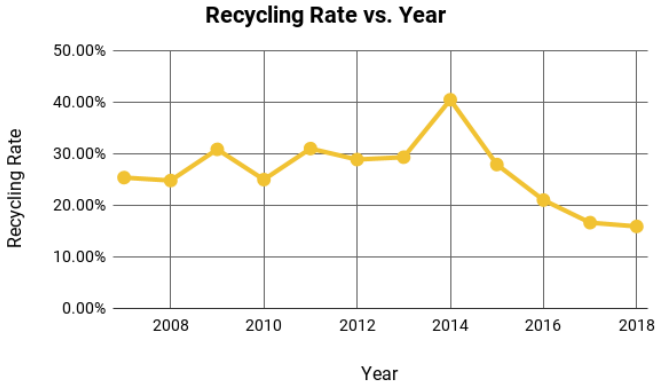


Figure 20: Recycling rate vs year

around 24 million tons of municipal waste are generated annually, according to the Egyptian Ministry of Environment.

3. Illiteracy

As shown in **Figure 14**, the percentage of learners in Egypt reached around 74.5%, meaning that around 25.5% of the population is illiterate. This high percentage of illiteracy is one of the major causes of Egypt's low recycling rate. People also are not interested in dumping garbage in places that allow the government to easily collect and recycle it, making recycling a difficult process for the government and garbage collectors.

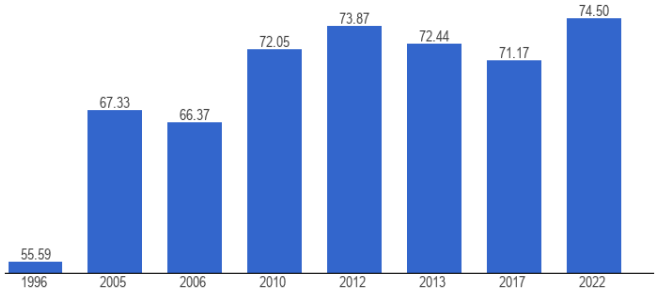


Figure 21: Percentage of learners over the years

Impacts:

1. Pollution

Egypt produces about 50-60 thousand tons of waste per day, or roughly 22 million tons per year. Only 60-74% of waste is collected; the remainder is disposed of in the streets, canals, informal dumps, or burned, causing a significant increase in the percentage of CO_2 and other pollutants in the atmosphere. According to official figures, less than 20% of collected waste is recycled; the rest is dumped or burned.

2. Health risks

A study of 275 municipal waste workers in Egypt showed significantly higher rates of respiratory symptoms, gastrointestinal issues, skin infections, musculoskeletal pain, and reduced lung function compared to controls. These

workers also had lower red blood cell counts, hemoglobin, hematocrit, and MCH, potentially due to chronic exposure to contamination. Scavengers in Mansoura had an alarming **43.3% prevalence** of hepatitis C antibodies, strongly linked to handling infectious waste. A known case involved a 55-year-old waste collector contracting hepatitis C from a discarded syringe.

3. Economic impacts

Collecting and disposing of solid waste currently costs around **EGP 125 (~USD 8)/ton** to collect and **EGP 100 (~USD 6.30)/ton** to dispose of, but fees from households generally cover only 20% of those costs. This creates a substantial funding gap, as municipalities must rely on subsidies and loans to maintain basic waste management. In 2018–19, Egypt spent just **EGP 4 billion (~USD 220 million)** on solid waste, part of a national budget of **EGP 1.424 trillion**, highlighting chronic underfunding (ratio of 1:356), compared to e.g., NYC's 1:52 ratio.

Problem to be solved

Increasing the capabilities of research in Egypt's laboratories through recycling electronic waste

Electronic waste, defined as any electronic appliance that is considered to have reached the end of its usefulness, has been rapidly increasing in the Egyptian landfill sites & markets. According to GEWSP (The Global E-Waste Statistics Partnership) electronic waste in the Egyptian market (EEE) has

increased from 754.0 kt in 2015 to 1069.0 kt in 2019. As e waste generated increased from 477.0 kt in 2015 to 586.0 kt in 2019 & e waste generated (kg per capita) increased from 5.16 kg to 5.85 kg. Additionally, the said increase in electronic waste **is reflected in Figure 15** through Egypt's rank in electronic waste generation among African countries.

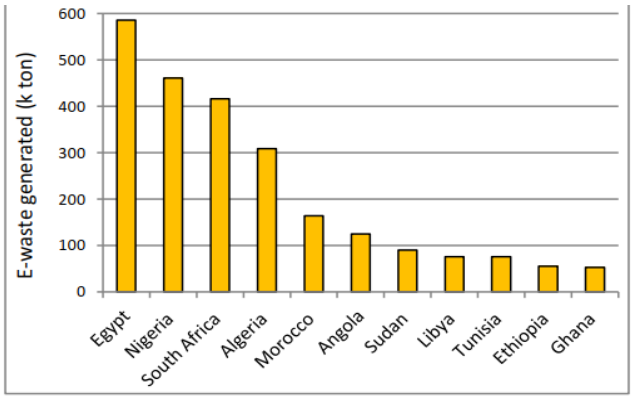


Figure 22: Top 10 countries in Africa generating e-waste (k-ton) (2019). Source: The Global E-waste monitor 2020

Consequently, the increase in EEE put on the market is almost 3.6 times the increase in population, while the increase in e-waste generated is almost double the increase in population. In average, the e-waste generated is almost 58.7 % of the amount of EEE put on the market. The e-waste generated per capita is 5.84kg/capita at an increasing rate of 2.66%. The e-waste generated in Egypt is about 20.1% that of the African continent, while the population in Egypt is about 8.6% of Africa. Previous statements in addition to the lack of any official large-scale projects to recycle electronic waste led to tons of precious materials such as gold, silver, palladium and indium to be wasted. As the waste is instead handled by local garbage traders & collectors.

Positive Consequences

1- Lower Precious Metal Prices

Electronic waste contains precious metals that could be extracted in specialized sites and be reused in other industries or sold directly to the market. According to a research paper published by OR Spectrum, electronic waste is typically compromised by 20.1% Copper, 8.1% Iron, 4.0% Tin, 2.0% Nickel, 2.0% Lead, 2.0% Aluminum, 2.0% Silver, 1.0% Zinc, 0.1% Gold, and trace amounts of palladium.

However, these materials are seldom extracted due to the lack of any official large scale e-waste recycling projects, thus Egypt loses tons of precious materials of significant values yearly.

Utilizing electronic waste recycling sites would greatly reduce the price of these metals by increasing the production and supply of precious metals.

2- Reduce Pollution

586.0 kt of electronic waste is disposed off in Egypt that gradually end up in landfill sites and garbage traders. Parts of these electronics are scrapped for iron and copper by local garbage recyclers that do not have the capacity to recycle the waste to its fullest potential. As only 20% of the e-waste is properly recycled and the remaining 80% is driven into uncontrolled landfills.

Furthermore, this influx of electronic waste into landfill sites contributes to the municipal solid waste as MSW production has reached 0.3 to 0.8 kg//capita as stated by the Egyptian Environmental Affairs Agency (EEAA), which increases yearly at rate of 3.4%. In addition to that, more than 60% of electronic waste is non-metal materials which are often disposed of in the environment by the local garbage traders and recyclers in addition to metals that aren't profitable to recycle in a small scale, which increases plastic & heavy metal poisoning.

Therefore, deploying country-wide electronic waste management sites would contribute to reducing pollution in Egypt through reduction of the MSW increase, uncontrolled garbage traders, and plastics poisoning.

3- Aid Innovation Increase

Egyptian research institution and universities have a lack of research equipment such as computers & laboratory machines. As a result, Egypt lacks in innovation compared to other countries of the region, as according to ECASTI Egypt has a GII level of 108 out of 142 countries where only four Arab countries are ranked lower than Egypt on the GII: Syria, Algeria, Sudan and Yemen.

Utilizing electronic waste to produce said equipment would reduce the costs of elevating the innovation of Egypt. The waste could be scrapped for raw materials or parts that are then assembled into new, functional research equipment.

Negative Consequences

1- Water and Soil Intoxication

Untreated electronic waste allocated to uncontrolled landfill sites and illegal garbage traders and recyclers pose a significant threat to soil and water.

As untreated electronic waste typically have 60.4% of non-metal materials which a local garbage trader cannot treat, instead burning in a landfill or throwing them in a body of water.

Furthermore, some components from electronic waste such as PVC plastics release dangerous gases such as hydrochloric acid into the atmosphere when burned which form acidic rain and intoxicating the air.

Other components such as LCD panels contain poisonous metals such as mercury which are released into bodies of water, infecting wildlife and soil.

2- Spread of Uncontrolled Recyclers

Uncontrolled recyclers and garbage traders in Egypt are considered a widespread issue in the face of the sanitation industry, due to their unregulated methods of recycling. For instance, non-metal materials are overseen by recyclers and are either burned or thrown into uncontrolled landfill sites.

Rapid increase of electronic waste fuels such industry by providing sources of copper, iron and aluminum, which typically make 30.2% of the waste. The collected materials are then either sold or processed in low-tech environments, where the rest of the waste components get burned and released into the environment.

3- Revenue Loss

Globally, USD 91 billion is wasted every year through electronic waste, which reflects the cruciality of electronic waste recycling sites. Yearly, 586.0 kt of electronic waste is generated in Egypt, only 20% of which is recycled while 80% are lost in landfills & illegal recyclers.

Typical electronical waste components are composed of 0.1% gold, which makes up for 88% of income from electronic waste. Additionally, the waste typically contains 0.2% Silver and 20.1% copper, which are all profitable metals, not considering iron and non-metal components also able to be recycled for profit.

With that in mind, only 20% of this waste is properly recycled. Which means millions of possible revenues are lost yearly. Including job opportunities, innovation advancement, environmental improvement, and industrial development.

Research

Topics related to the problem:

1. Environmental Pollution.

Since climate change and pollution are the main points in the challenge, researching them was essential for defining the problem. Environmental pollution splits into subcategories such as air, water, and soil pollution, and they are mostly caused by Electronic, Garbage Waste, and gas emissions.

While holding the 12th position in the global pollution index, Egypt is also ranked 9th in terms of the highest air pollution globally. According to the pollution index, Cairo scores 95.8, with the annual mean concentration of fine particulate matter ($PM_{2.5}$) in Egypt being around 72 $\mu g/m^3$, 10 times above the safe limit.

2. Management of Electronic Waste.

Recycling Electronic Waste is a critical part of the challenge, where the electronic waste generated in Egypt is one of the biggest causes of pollution and climate change, making it important to study and research.

According to the EEAA (Egyptian Environmental Affairs Agency), the annual electronic waste generation in Egypt surpassed 586 kilotons in 2009, calculating up to 5.84 KG per capita, an atrocious number for one person, especially since only 10% of the waste is collected and recycled; moreover, 1069 KT are electrical or electronic equipment, which can be used for many projects.

3. Research Capacity.

The research capacity is the focus and target of the challenge, where the goal is to increase it to collaborate with the research and development, so

studying Egypt's research capacity is a fundamental step to identify the main problem.

Egypt spends only around 1.03% of its GDP on research and development, **according to a 2021 science report by UNESCO**, which is way below the global average of 2.5%; showing that research in Egypt is well underdeveloped; moreover, Egypt has limited research infrastructure, where most of the universities and research centers lack modern laboratory facilities and equipment, especially outside of Cairo and Alexanderia, where research facilities are concentrated, causing regional inequalities. The collaboration between universities and industry is also weak as only 10% of research projects are commercialized.

Topics related to the solution:

1. Importance of bacteria.

Bacteria have many crucial roles in a lot of fields. Environmentally, bacteria break down organic waste, collaborating with the nutrient cycle, and bacteria such as methanogenic and photosynthetic bacteria, contributing to the renewable energy production and Oxygen production, respectively. In the medical field, it's so important in medical production. For instance, Bacteria like *Streptomyces* produce antibiotics such as streptomycin, tetracycline, erythromycin, and Non-pathogenic bacteria are used as carriers or vectors in vaccine production.

In conclusion, bacteria are crucial in life, which is why they're an important part of the solution to the challenge.

2.Thermal-Related Electronic Waste.

Among the generated electronic waste in Egypt, many thermal-related waste items are found to still function well enough to be recycled into new operations. This was helpful in the sake of deciding to make the solution thermal-related.

From the total of 690 kilotons of generated electronic waste in Egypt, household appliances make the major share, including thermal-related electronic waste such as refrigerators, water heaters, AC units, and

microwaves. They represent a big slice of sales, which is why they generate a lot of mass when they reach the end of use. Consumers usually repair or resell these wastes. This culture of reusing or reselling means that these wastes still work.

3. Insulation systems.

Insulation systems were a crucial part of the solution to insulate the temperature and humidity inside the project, which mitigates bacteria. There are many insulation materials.

An insulating material has low thermal conductivity, such as fiberglass, polyurethane foam, and mineral wool. Fiber glass has a thermal conductivity of only 0.035 W/m.K, with a temperature range of 540 °C. It is used in the walls of ovens and refrigerators. Polyurethane foam's thermal conductivity is around 0.020 W/m.K, making it an excellent insulator, where it's mostly used in refrigerators and water heaters.

The prior solutions

OM118-M3 40X-400X Monocular Student Compound Microscope

The compound microscope is widely thought to have been created by Zacharias Janssen, using collapsing tubes that produced magnifications up to 9X. However, most experts believe that his father, Hans, must have had a significant hand in the construction of the instrument. A compound microscope consists of several key components that can magnify small objects or specimens. Understanding these parts is essential for

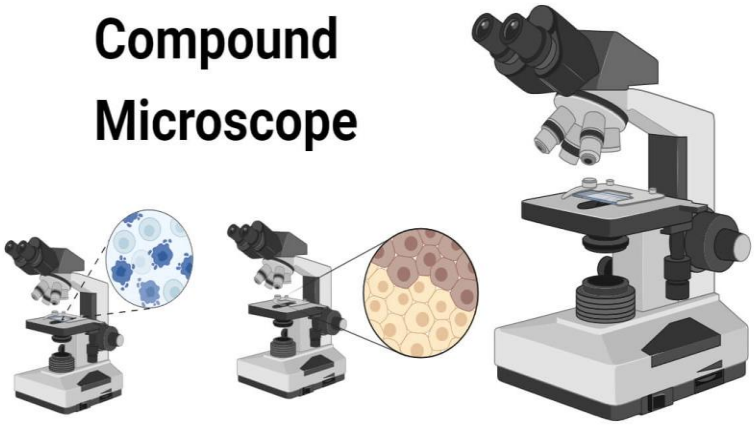


Fig 23: compound microscope

knowing how a compound microscope works. These components are, firstly, the **eyepiece lens,** which is the lens that you look through to see the image of the specimen. It is usually located at the top of the microscope. Secondly, **the Objective lens** is our lens that is located at the bottom of the microscope, near the specimen. Revolve on a rotating nosepiece. And each objective lens has a different magnification power, typically ranging from 4x to 100x. The third part is the **Stage:** The stage is the platform on which the specimen is placed. It is usually located in the middle of the microscope.

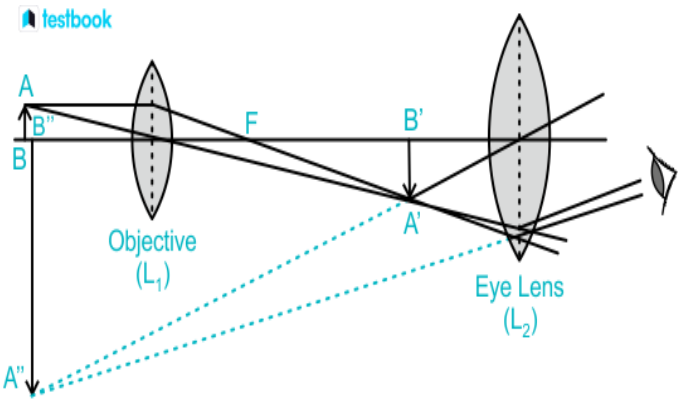
Light source: The light source is the source of light that illuminates the specimen. It is usually located at the bottom of the microscope.

Diaphragm: The diaphragm is a device that controls the amount of light that reaches the specimen. It is usually located below the stage.

Focusing knobs: The focusing knobs are used to focus on the image of the specimen. They are usually located on the side of the microscope.

Mechanism

The light source in the microscope's base emits a beam of light that is then concentrated and directed at the specimen by a condenser lens. Thereafter, the transparent glass slide with the specimen is placed on the stage and positioned between the condenser and the objective lens. The objective lens, located closest to the specimen, collects the light and forms a real, inverted, and magnified primary image within the body tube. This event occurs when the specimen is placed just outside the focal length of the objective lens. Additionally, the primary image serves as the "object" for the eyepiece, which is also known as the ocular lens. Next, the eyepiece magnifies this real image a second time to produce a virtual, enlarged, and inverted image that the user sees. Then, a fine-adjustment knob moves the stage to bring the image into sharp focus, with the result viewed as a virtual image, either at the user's near point of vision or at infinity. For the last thing, the objective lens



generates a true, inverted picture, but the eyepiece functions as a basic magnifier that does not invert and generates a virtual image. As a result, the picture is inverted and virtual overall. Additionally, for the calculation of the magnification power $M = m_o * m_e$, where,

M = magnification power of the compound microscope

m_o = magnification power of the objective lens

m_e = magnification power of the eyepiece

For instance, when the objective lens can magnify by 40 times and the ocular by 10 times, the specimen will be overall magnified by 400X.

When the final image is formed at infinity, it is given as:

$$\mathbf{M = D/ f_o * L/f_e}$$

where,

D = Least distance of distinct vision = 25cm

L = length of the microscope tube

f_o =focal length of objective lens

f_e = focal length of eyepiece

points of strength

1. Ease of use:

Compound microscopes are user-friendly due to features like being built with a source of lighting inside them, simple slide placement, and coarse and fine focus on manual adjustments. However, ease of use depends on the model's design, and proper technique is needed to avoid damage. Key steps for ease of use include starting with the lowest magnification power, properly adjusting the light to have the best experience, and using the focus knobs correctly. Also, due to their straightforward single-eyepiece design, monocular microscopes are user-friendly and ideal for beginners learning

microscopy techniques, so they can move to a more complex model in the future.

2. Portable and Compact

The compound microscope has been proven to be lightweight, durable, so it can be carried easily in different locations and features a robust metal construction, unlike the devices that can also be made from e-waste, like the incubator and the magnetic stirrer and Many models of the compound microscope have an LED source for specimen illumination and which is not a feature in other devices if you want to use them at the night, in addition of a insane feature that is digital connectivity and this with some advanced models of the compound microscope you can connect the smartphones and tablets to view the samples in a bigger and more detailed screen and can take photos, videos for samples

3. Sufficient for Basic Applications:

Unlike the other devices that can be made from E-waste (as an incubator and magnetic stirrer that are used in biology or physics only), the compound microscope (generally microscopes) can be used in many applications, like biology in studying the structure, cells, tissue, and microorganisms. Second, in the medical field, it is used to assist in pathology labs by allowing doctors to analyze blood and cell samples to diagnose diseases. Next, in education, this makes for the students a good intro to microscopy and observing specimens in a laboratory, and lastly, in forensics, in examining the evidence, such as fingerprints, hair, or fibers, in a crime

The points of weakness

1- Eye Fatigue:

Compound microscopes, when used for a lengthy amount of time without proper adjustment, eye fatigue and soreness can easily occur, as the

eye muscles continually work against a poorly adjusted instrument as the eye scans the
image. As a result, the lesser adjustment of the compound microscope, which is part of its simplicity, amplifies such reactions in the eyes. Furthermore, inadequate lighting is often present in compound microscopes, combined with the necessity to keep an eyes closed for a long time, which increases eye fatigue.

2. Complexity in Usage

Operating a compound mild microscope requires a significant studying curve as compared to easy magnifiers. It isn't always a "factor-and-shoot" device; accomplishing a extremely good picture calls for getting to know unique optical alignment techniques, such as Köhler illumination. This technique by myself involves more or less 7 to ten awesome steps, consisting of centering the light supply, focusing the condenser, and adjusting the field diaphragm and aperture iris to stability evaluation and determination. Furthermore, sample instruction is notably technical; specimens ought to be sliced to a thickness of about five to 10 micrometers (kind of the scale of a unmarried red blood cellular) to allow light to pass through. Educational resources from universities frequently note that students require several hours of hands-on schooling to grasp 100x oil immersion strategies without destroying goal lenses, which may cost between $500 and $2,000 for high-grade research models.

3. Limited Depth of Field

The maximum profound optical trouble of a compound microscope is the extremely shallow depth of discipline (the vertical distance of the specimen that remains in focus at one time). There is an inverse relationship between magnification and the depth of the subject. According to optical microscopy requirements, at the same time as a 4x goal may provide an attainable intensity of area, a strength 100x objective (general magnification of 1000x) reduces the intensity of discipline to about 0.2 to zero. Four micrometers. Since many biological cells are 10 to 30 micrometers thick, a consumer can by no means see the complete cell in consciousness concurrently. Instead, they see a skinny optical "slice." To view the whole specimen, the consumer ought to continuously modify the satisfactory recognition knob to visually experiment

thru dozens of focal planes, making it impossible to capture a real three-dimensional image in a single snapshot without specialized software stacking.

VEVOR 20L lab incubator

As laboratories expand in numbers and scope, there is a growing demand for affordable and compact equipment capable of maintaining controlled environments for biological and microbiological work. The VEVOR 20L laboratory incubator was designed to meet this need, providing a cost-effective solution for bacterial culture, seed germination, and general laboratory heating applications.



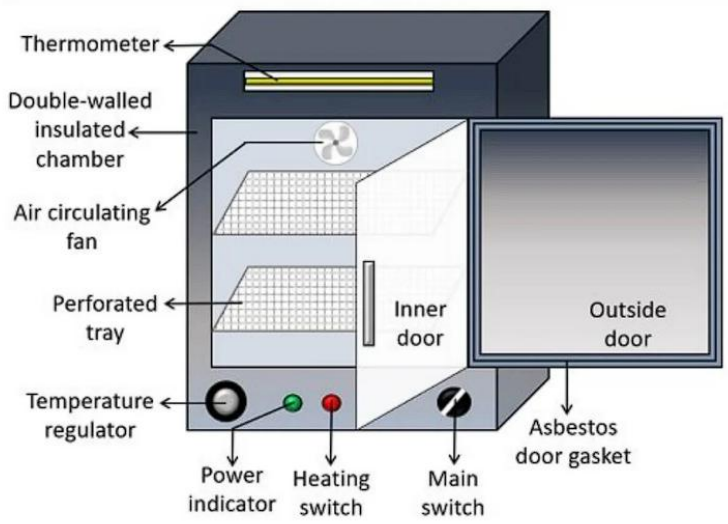
Figure 24: VEVOR 20L Bacteriological Laboratory Incubator

Launched by the VEVOR Chinese company, this incubator model, priced at $253, typically features a chamber measuring 260 * 280 * 270 mm and has a total capacity of 20 liters. It has an outer shell made of painted or powder-coated steel and an interior of SUS201 stainless steel, making it highly resistant to corrosion from humidity or other causes. This model is a dry-heat type, meaning that it does not have features like humidity control, cooling, or CO_2, causing it to be one of the low-priced models on the market at the time, **as shown in Figure** 16.

This incubator is usually used for incubating medium-sized Mesophilic bacteria at temperatures stable at 30-37 °C, being the optimal temperature for growing and researching types like E. coli and Bacillus. To provide such temperatures, the incubator needs a power supply of 110V or 220V. It comes with a temperature range, being the room temperature + 5 °C → ≈ 65–70 °C, and a temperature accuracy of ±1–2 °C (typical spec). To insulate that heat on the inside, it features a heat-resistant foam layer and a silicone door seal. Finally, Italy has a tempered glass pane in the front door to make it see-

through, so the researcher would always be aware of any sudden changes in the specimen.

Mechanism



While most of the incubator types give access to the user to change the humidity and CO_2 levels, this model is called a dry-heat incubator, primarily designed for non-pathogenic bacterial incubation, meant for types that don't require anything more than a stable temperature to be preserved and grown at a stable rate, like E. coli and Bacillus. This model relies on an electrical heating system to provide the needed heat for the specimen on the inside, a temperature sensing and feedback control to check up on the system to know if there are any errors, and an air circulation and convection system to maintain an equal temperature in every point on the inside to ensure an even growth rate for the sample, and a thermal insulation and enclosure design to minimize the heat loss in the interior. These systems work together based on key physical and engineering **principles**, mainly **thermodynamics**, **heat transfer**, and **feedback control theory**.

The incubator functions as a closed thermodynamic system that maintains a steady-state temperature above the surrounding environment. According to the **First Law of Thermodynamics**, the electrical energy supplied to the heating element is converted into internal thermal energy inside the chamber. The balance between **heat input (Q_1)** and **heat loss (Q_2)** determines the chamber temperature: $Q_{in} = Q_{loss} + Q_{stored}$. At equilibrium, when $Q_{in} = Q_{loss}$, the temperature on the inside remains constant. The insulation, chamber material, and ambient conditions all affect this energy balance. To transfer heat, three

different types of heat transfer occur inside the incubator. The first type is conduction, in which Heat moves through the metal walls and insulation by molecular vibration. The stainless-steel chamber conducts heat efficiently, helping stabilize the temperature throughout the inner surface. The second type is convection, in which the internal fan creates **forced air convection**, circulating warm air uniformly. Convection is the main mechanism of distributing heat to all parts of the chamber, ensuring samples receive similar conditions. The last mechanism of heat transfer in the system is radiation, in which heat radiates from the heating element and metal walls to other surfaces inside. Although minor, it helps maintain uniform temperature between the walls and the air.

Points of strength

1.Low cost and high accessibility

Due to its simple design and abundance in the market, this design has an incredibly low cost compared to other designs in the market at the current time. It is usually sold for $250-$400, while the other models on the other hand are priced at $400+. Because of this relatively low price, this incubator makes incubation accessible to schools, training laboratories, and small research projects. It provides good functionality for its price, covering essential incubation needs for starter labs.

2.Compact and lightweight design

This design is widely known for its excellence in terms of design compactness, as it has a maximum capacity of holding 20L at a time. It is considered one of the most compact designs that one can find in the same price bracket. It is also known for being one of the most lightweight incubator designs that can even be found, weighing only 6kg-8kg. An incubator being this lightweight and compact can be effortlessly placed on benchtops or within teaching laboratory setups in schools, university labs, or small research labs.

3.Energy efficiency

The VEVOR 20L incubator is considered incredibly energy-efficient compared to other models in the same class because of its small chamber

volume, moderate heating power, and basic insulation design. It consumes significantly less electricity than larger or multifunctional laboratory incubators, like CO_2 or refrigerated models, while still maintaining stable temperatures between 25 °C and 65 °C. The power consumption heavily depends on the operating phase; for instance, it consumes 400W in the warm-up phase, requires an average of 100-200W in the steady phase, 5-20W in the idle state (power on, heater off), and when the door is opened, temperature drops, making the controller increase the heater output temporarily, driving it to its power consumption peak.

Points of weakness

1. Temperature inaccuracy

A huge flaw in the design is that the incubator's proportional-integral-derivative (PID) controller and sensor calibration are not highly precise. Actual chamber temperature can differ from the displayed value by **±2-5 °C**, especially near the edges or door. This variation can affect the bacteria's growth, making it grow unevenly or at a non-uniform rate and affecting experimental reliability. The cause of this severe flaw is that the incubator relies on a single low-cost thermistor and lack of factory calibration or certification of the PID controller.

2. Fan noise and vibration

Cheap fans like the one used in the motor often have a slight imbalance on the blades, causing the incubator to sometimes vibrate or make a noticeable noise measured to be around 45-50 dB. The use of poor bearings for the motor also causes it to vibrate even more than the fan blades' imbalance. The constant noise caused by the incubator can often affect the laboratory's research or distract the students' activities. It also might affect the growth rate or the shape of the colony, as the vibrations often disturb the airflow inside.

3.Limited lifespan

The VEVOR 20L incubator usually lasts for around 2-3 years of frequent use. Compared to professional laboratory incubators that can last up to 10 years of constant use, this incubator's lifespan is considered extremely short. This happens due to the constant expansion and contraction of the

resistive heating element every time it heats up and cools down. Over thousands of constant heating and cooling cycles, thermal fatigue forms microcracks and oxidization on the wire surface, reducing the efficiency and making the heat spread unevenly, eventually maki

Thermo Scientific Heratherm IGS180

To match the rapid expansion of professional laboratory needs, the current technologies are supposed to become more and more advanced as well. The Thermo Scientific Heratherm IGS180 is an incubator uniquely made for heavy-duty duties in professional research laboratories. Compared to other incubators within the same price range, the IGS180 distinguishes itself through its exceptional thermal stability, energy efficiency, and build quality, delivering performance that fully justifies its position among the leading models on today's market. Its excellence in maintaining a stable environment to grow various types of bacteria due to its high temperature range makes it an optimal choice for advanced research laboratories.



Figure 25: Thermo Scientific Heratherm IGS180 Incubator

Launched by Thermo Fischer Scientific, this incubator model, priced at $3250-$4490, typically features a chamber measuring 835 mm x 600 mm x 700 mm and has a total capacity of 194 liters. It features an outer shell made of painted or powder-coated double-layer steel for mechanical protection and an interior constructed from AISI 430 stainless steel, making it easy to clean and highly resistant to corrosion from humidity and other environmental factors. This model is a dry-heat type, meaning that it does not have features like humidity control, cooling, or CO_2.

This incubator is usually used for incubating medium-sized Mesophilic and Thermophilic bacteria at temperatures stable at 30-37 °C, being the optimal temperature for growing and researching types like E. coli and Bacillus. To provide such temperatures, the incubator needs a power supply of 120V, 60Hz or 230V, 50Hz/60Hz. It comes with a temperature range, being the room temperature +5 °C → ≈ 70 °C, and a temperature accuracy of ±0.2 °C (typical spec). To insulate that heat on the inside, it features a heat-resistant foam layer and a high-quality silicone door gasket. It finally has a double-glazed viewing glass window, made of two parallel layers of tempered glass with an air gap between them, so the researcher would always be aware of any sudden changes in the specimen.

Mechanism

While most of the incubator types give access to the user to change the humidity and CO_2 levels, this model is called a dry-heat incubator, primarily designed for non-pathogenic bacterial incubation, meant for types that don't require anything more than a stable temperature to be preserved and grown at a stable rate, like E. coli and Bacillus. This model relies on an electrical heating system to provide the needed heat for the specimen on the inside, a temperature sensing and feedback control to check up on the system to know if there are any errors, and an air circulation and convection system to maintain an equal temperature in every point on the inside to ensure an even growth rate for the sample, and a thermal insulation and enclosure design to minimize the heat loss in the interior. These systems work together based on key physical and engineering **principles**, mainly **thermodynamics**, **heat transfer**, and **feedback control theory**.

The incubator functions as a closed thermodynamic system that maintains a steady-state temperature above the surrounding environment. According to the **First Law of Thermodynamics**, the electrical energy supplied to the heating element is converted into internal thermal energy inside the chamber. The balance between **heat input (Q_1)** and **heat loss (Q_2)** determines the chamber temperature: $Q_{in} = Q_{loss} + Q_{stored}$. At equilibrium, when $Q_{in} = Q_{loss}$, the temperature on the inside remains constant. The insulation, chamber material, and ambient conditions all

affect this energy balance. To transfer the heat, three different types of heat transfer occur inside the incubator. The first type is conduction, in which Heat moves through the metal walls and insulation by molecular vibration. The stainless-steel chamber conducts heat efficiently, helping stabilize the temperature throughout the inner surface. The second type is convection, in which the internal fan creates **forced air convection**, circulating warm air uniformly. Convection is the main mechanism for distributing heat to all parts of the chamber, ensuring samples receive similar conditions. The last mechanism of heat transfer in the system is radiation, in which heat radiates from the heating element and metal walls to other surfaces inside. Although minor, it helps maintain a uniform temperature between the walls and the air.

Points of strength

1.Energy efficiency

The IGS180 incubator is considered incredibly energy-efficient compared to other models in the same class because of its moderate heating power and basic insulation design. It consumes significantly less electricity than larger or multifunctional laboratory incubators, like CO_2 or refrigerated models, while still maintaining stable temperatures between 30 °C and 70 °C. The power consumption heavily depends on the operating phase; for instance, it consumes 400W in the warm-up phase, requires an average of 100-200W in the steady phase, 5-20W in the idle state (power on, heater off), and when the door is opened, temperature drops, making the controller increase the heater output temporarily, driving it to its power consumption peak.

2.Durable construction

The **Thermo Scientific Heratherm IGS180** is built for long-term durability and reliability in demanding laboratory environments. Its **stainless-steel inner chamber** resists corrosion and allows easy cleaning, while the **powder-coated steel exterior** protects against mechanical damage. A layer of **polyurethane foam insulation** between the walls enhances both strength

and thermal efficiency. The **double-walled design** and **silicone door gasket** ensure minimal heat loss and consistent internal conditions, giving the incubator a lifespan that often exceeds ten years of continuous use.

3. silent operation

The **Thermo Scientific Heratherm IGS180** operates with exceptional silence, making it ideal for laboratories that require a low-noise environment. Unlike fan-assisted incubators, it uses a **gravity convection system**, where warm air naturally circulates without mechanical airflow. This eliminates vibration and noise produced by internal fans, protecting sensitive cultures from mechanical disturbance and ensuring stable incubation conditions. The absence of moving air parts also reduces wear, contributing to the incubator's long service life and energy efficiency. As a result, the Heratherm IGS180 provides a quiet, stable, and reliable atmosphere for precise microbiological work.

Points of weakness

1. High initial cost

One first-rate disadvantage of the Thermo Scientific Heratherm IGS180 is its high initial cost, which limits its accessibility to smaller laboratories and educational establishments. Priced usually between $3,000 and $5,000. The incubator represents considerable funding in comparison to lower-capability or access-level fashions. This high charge is specifically due to its advanced PID manipulation system, long-lasting stainless-steel construction, and precision engineering, all of which ensure top-notch reliability and accuracy. While the value is justified by using its long lifespan and power-efficient operation, it may nonetheless be financially difficult for laboratories with restricted budgets or those requiring more than one incubation unit.

2. Heavy weight

The Thermo Scientific Heratherm IGS180 has a noticeably heavy weight, averaging between 60 and 65 kilograms, that may pose demanding situations in terms of mobility and setup. Its strong production—providing double metal partitions, dense polyurethane insulation, and a reinforced stainless steel chamber contributes to its durability and mass. While this strong construct enhances stability and thermal retention, it also makes the incubator difficult to relocate once installed. As a result, it's miles a great idea for everlasting placement on a strong laboratory bench or floor stand, where it is able to perform properly without the need for common movement.

3.Slow temperature recovery

A minor hindrance of the Thermo Scientific Heratherm IGS180 is its slow temperature recovery after door openings. Because the incubator uses a gravity convection machine rather than a fan-assisted airflow, the inner air movement relies entirely upon the natural movement of warm and cool air. This layout guarantees quiet, vibration-free operation but results in a quick postponement in restoring the set temperature as soon as the door is opened and heat escapes. Although this recuperation length is usually brief—only a few minutes—it can slightly have an effect on experiments that require common access to the chamber. However, the alternative is advanced strength performance and more pattern balance at some stage in undisturbed operation.

Ohaus Guardian 2000 Stirrer | e-G21ST04R

The Ohaus Guardian 2000 Stirrer is designed to deliver powerful stirring performance for laboratory programs. It offers a flexible speed variety from two hundred to 1600 rpm, ensuring thorough blending of materials throughout numerous viscosities and volumes. With a maximum capability of 15 liters, the stirrer comprises vast quantities of samples and answers, assembling diverse experimental needs. Featuring a resilient resin plate measuring 10.2 x 10.2 cm, the stirrer ensures uniform stirring motion and inexperienced warmth distribution. The sturdy steel casing of the e-G21ST04R offers balance at some stage in operation and offers durability in laboratory environments. For person safety and operational attention, the stirrer is ready with a hot top indicator light that signals customers to more than one floor temperature, ensuring secure management. Additionally, a stirring indicator mildly indicates the operational popularity of the stirrer, facilitating easy monitoring and management throughout the course of experiments. The manipulation of the stirrer is simple, with an unmarried analog knob, permitting clients to adjust stirring settings. The PET of the front panel complements durability and resilience to laboratory conditions, ensuring long-lasting performance. Additionally, the e-G21ST04R comes with a stir bar protected in PTFE, which allows substances to blend smoothly and successfully, main to steady and reliable results. These capabilities together make the Ohaus Guardian 2000 a reliable choice for precise stirring obligations in laboratory settings. And this magnetic stirrer has many functions: First, its speed varies are 2 hundred rpm to 1600 rpm; second, its maximum capacity is 15 L; 1/3, its resin plate is 10.2 x 10.2 cm; fourth, it has a heating top indicator light, a stirring indicator light.



Fig 26: Ohaus Guardian 2000 Stirrer | e-G21ST04R

Mechanism

The operating principle behind a magnetic stirrer lies in the interaction between magnetic fields. The device consists of a rotating magnet or an electromagnet. When turned on, the magnet under the platform begins to rotate, creating a rotating magnetic field. A stirring rod, usually made of inert material such as PTFE, with a small magnet internally, is positioned to align with this field to the north. As the arena rotates, it forces the stirring rod to rotate at a controlled speed. This rotation creates a vortex in the liquid, which improves mixing and ensures that everything from solute to suspension is evenly distributed. This principle makes the magnetic stirrer particularly reliable and safe. With no gears or shafts in direct contact with the north, the risk of infection is minimal. For superior models, functions inclusive of programmable settings, variable speed manipulation, and reversible motion enhance precision and performance. For example, the Magnetic Hot Plate Multi-Position Stirrer from Borosil Scientific uses a microprocessor-managed era to offer ordinary mixing with speeds ranging from 300 to 1500 RPM. These stirring characteristics are lengthy-lasting, chemical-resistant housing that withstands day-to-day laboratory use. The integrated support rod holder is compatible with different series additives, accommodating the protected experimental setups. As the magnetic situation spins, the stir bar spins with it, developing every day and smoothly blending the liquid inside the field. And from the blessings, on account that there's no direct bodily touch between the motor and the liquid, the method is low-maintenance, requires minimal cleaning, and is much less probably to introduce infection.

Points of strength

1. Affordability

Discover the Ohaus Guardian™ 2000 Stirrer e-G21ST04R, a dependable and affordable laboratory stirrer designed for reliable mixing in educational, research, and industrial labs. This robust unit offers a 15 L capacity with a 4 x 4-inch resin plate, stirring up to 1,600 rpm. The analog

knob provides easy adjustment of stirring speed, ensuring user-friendly operation, a durable metal housing, and a PTFE-coated stir bar. The e-G21ST04R is the ideal solution for labs seeking efficient and precise sample mixing.

2. stirrer performance

The Ohaus Guardian 2000 Stirrer is designed to deliver effective stirring performance for laboratory applications. It offers a versatile speed range from 200 to 1600 rpm, ensuring thorough mixing of substances across various viscosities and volumes. With a maximum capacity of 15 liters, the stirrer accommodates substantial quantities of samples and solutions, meeting diverse experimental needs.

Featuring a resilient resin plate measuring 10.2 x 10.2 cm, the stirrer ensures uniform stirring action and efficient heat distribution

3. clear visibility

The shiny white floor of the magnetic stirrer is a crucial energy layout. It provides great visual comparison, permitting clear commentary of pattern coloration changes, reaction progress, and vortex formation. This direct monitoring functionality is emphasized via enterprise leaders, together with Ikea and Corning, which specify bright panels for the highest quality visibility. By permitting researchers to as it should be song experiments in real time, this selection will increase accuracy, improve management, and contribute to safe laboratory operation by means of allowing any sudden changes to be detected right away.

Points of weakness

1. Weak Magnetic Coupling

The core of a magnetic stirrer's feature is the invisible "grip" among the motor's rotating magnet and the stir bar. This coupling weakens substantially with distance. If the vessel base is overly thick or the liquid quantity is too high, the elevated pole spacing forces the motor to work more difficult, regularly resulting in slow bar movement or complete failure to spin. Similarly, a poorly designed plate magnet or an underpowered motor can't maintain a strong magnetic area, especially under load. Ensuring a minimal

gap between the stir bar and the pressure magnet is the primary crucial step for dependable operation.

2. Motor Speed Inconsistency

A magnetic stirrer's motor is engineered for a particular torque variety. When tasked with blending dense or fairly viscous answers, or while driven to its maximum pace, the motor can warfare. This overload reasons the inner additives to work inefficiently, main to audible velocity fluctuations, inconsistent RPMs, or a entire stall. This inconsistency compromises experimental reproducibility and combining efficiency. For annoying packages, a stirrer with a effective, digitally managed motor that offers regular torque across its pace variety is crucial to keep a steady vortex.

3. Stir Bar Decoupling

The maximum common seen failure is decoupling, in which the stir bar sporadically "skips" or stops completely. This is typically resulting from a mismatch between the bar, the solution, and the set pace. A bar that is too small lacks the surface vicinity to grip the liquid, whilst an outsized bar creates immoderate drag. In viscous fluids, this drag overwhelms the magnetic force. The result is a chaotic spin-and-slip cycle that gives terrible mixing. Selecting an accurately sized bar for the vessel and viscos

Chapter 2

Solution requirements

Cost-Effectiveness

Cost-effectiveness refers to whether a product offers good features and high functionality compared to the cost of buying the product. Small-sized bacteriological incubators used in small laboratories and schools are considered cost-effective, due to the immense number of features provided for their price. They are usually priced at $200-$1000 on today's market.

Eco-Friendliness

A point that adds to a solution's strengths is being eco-friendly. Eco-friendliness in general refers to the quality of being environmentally sustainable or having a minimal negative impact on the natural environment. In scientific and industrial contexts, it means designing a solution in ways that reduce energy consumption, limit waste generation, and conserve natural resources.

Safety and reliability

Safety and reliability are essential characteristics of successful, widely used solutions. Especially in laboratory environments, bacteriological incubators must maintain precise temperature control without exposing the user or the samples to risk. Modern incubators often come with over-temperature protection, alarms, and automatic power cut-off systems to prevent accidents and sample loss. Durable construction materials, stable sensors, and accurate controllers ensure consistent operation and safeguard both users and experiments.

Design requirements

We will build up a single scientific prototype instrument used in at least one type of laboratory, whether it's Biology, Chemistry, Earth Science, Physics, or Fabrication Labs, and if the chosen scientific instrument possesses only one measurable function, we are required to present outcomes from two actual laboratory applications. Conversely, if the instrument has multiple measurable functions, at least one real laboratory application must be demonstrated.

The prototype must include an additional function compatible with the instrument's primary function, and the prototype should incorporate multiple components sourced from old, used, or discarded electrical devices. At least one must be an electrical component, while the remaining components will be chosen by the team based on the specific functional application of the selected instrument. Additionally, several items may be chosen from a different component category. And if the prototype includes sensors, they must be calibrated using standard methods or instruments.

The internal and external housing/cover of the prototype instrument should be made from a transparent material that may be extracted from the old devices, manufactured, or purchased. Additionally, the prototype's primary characteristics should be demonstrated, which include the necessary voltage, the maximum drag electric current, and the maximum operating power. In addition, other selected characteristics/requirements should be presented based on the instrument's functions.

A schematic diagram that illustrates both the external and internal structural design of the prototype instrument should be presented. Additionally, it should include the electric circuit diagram if utilized in the design or necessary for the operation of the prototype instrument.

Results of the prototype instrument trial test and its application/s should be presented using graphical fitting methods with at least five readings.--- The team should provide a video for the components' extraction, prototype building, trial tests, prototype characterization, and the instrument

application/s. The prototype's accuracy should be less than◌. +-5% of uncertainty.

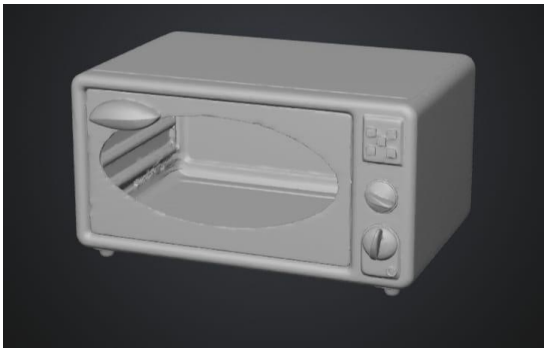
The selection of the solution

The solution chosen is an incubator. The thermal insulation prevents the outer temperature from affecting the incubator, where the Arduino takes the temperature readings from the thermostat and accordingly controls the thermal core to achieve the desired circumstances for the targeted cells, such as bacteria, yeast, and mammalian cells, which have specific conditions for their metabolic and reproductive processes. While most of the incubator types give access to the user to change the humidity and CO_2 levels, this model is called a dry-heat incubator, primarily designed for non-pathogenic bacterial incubation, meant for types that don't require anything more than a stable temperature to be preserved and grown at a stable rate, like E. coli and Bacillus. This model relies on an electrical heating system to provide the needed heat for the specimen on the inside, a temperature sensing and feedback control to check up on the system to know if there are any errors, an air circulation and convection system to maintain an equal temperature at every point on the inside to ensure an even growth rate for the sample, and a thermal insulation and enclosure design to minimize the heat loss in the interior. These systems work together based on key physical and engineering **principles**, mainly **thermodynamics**, **heat transfer**, and **feedback control theory**. The incubator functions as a closed thermodynamic system that maintains a steady-state temperature above the surrounding environment. According to the **First Law of Thermodynamics**, the electrical energy supplied to the heating element is converted into internal thermal energy inside the chamber. The balance between **heat input (Q_1)** and **heat loss (Q_2)** determines the chamber temperature: Qin = Qloss + Qstored. At equilibrium, when Qin = Qloss, the temperature on the inside remains constant. The insulation, chamber material, and ambient conditions all affect this energy balance. To transfer heat, three different types of heat transfer occur inside the incubator. The first type is conduction, in which heat moves through the metal walls and insulation via molecular vibration. The stainless-steel chamber conducts heat efficiently, helping stabilize the temperature throughout the inner surface. The second type is convection, in which the internal fan creates

forced air convection, circulating warm air uniformly. Convection is the main mechanism for distributing heat to all parts of the chamber, ensuring samples receive similar conditions. The last mechanism of heat transfer in the system is radiation, in which heat radiates from the heating element and metal walls to other surfaces inside. Although minor, it helps maintain a uniform temperature between the walls and the air.

Selection of Prototype

The selected prototype is a compact lab bacterial incubator, which will be used in school labs. This semester's challenge requires that all materials be extracted from electronic & electric waste. Therefore, all the materials used in the prototype will be extracted from said sources. These chosen sources for the prototype have been selected to be broken combination of ovens, microwaves, and/or traditional ovens. The stated appliances will then be scrapped for the prototype components. The most notable of said components is the thermally insulative chambers used in the combination ovens. These chambers are necessary to keep the heat always generated, insulated, and constant. Another main component would be the heat fans & thermostat. The fans help distribute the heat across the interior while the thermostat detects the temperature for the electronic controls. For the electronic system, the microwave or combination oven LED display will be rewired and chipsets from the original appliances into new the new system. Aside from that, transparent materials extracted from other appliances can withstand cutting, bending, and shaping, such as acrylic or transparent plastic will be used for the door or side windows of the prototype. These windows are crucial for observing the bacterial growth process during the incubation. The dimensions were selected based on the requirements of school laboratories and the limitations of the compact size and recycled components. The internal width is 32 cm, while the outer width is 45 cm. The internal height is 22 cm, while the outer height is 35 cm. The internal depth is 30 cm, while the outer depth is 40 cm. These selected



dimensions are based on the requirements and the limitations of a compact school laboratory incubator and the necessity of recycling electronic and electric waste.

Chapter 3

The abstract

Egypt encounters multiple challenges that hinder its growth towards self-independence. The project aims to solve the widespread e-waste crisis, promote recycling and sustainability, and provide an affordable alternative for school labs. This will be done through solving Egypt's grand challenges: improving the scientific and technological environment for all, reducing and adapting to the effects of climate change, recycling garbage and waste for economic and environmental purposes, and addressing and reducing pollution fouling our air, water, and soil. The purpose of the study is to utilize the abundant, unused electronic waste by turning it into a fully functional bacteriological incubator capable of serving in school labs. The selected problem to be solved is to increase the capabilities of research in Egypt's laboratories through recycling electronic waste. The device was constructed by recycling electronics from old devices, such as a PTC heater and a TEC cooler. Various tests have been done to ensure that the calculation error percentage does not exceed ±5%. There were some negative results overcome by different modifications. Major findings were the device's ability to generate temperatures ranging from 27 °C and 45 °C under average usage circumstances, with the ability to go beyond this range, if necessary, while major conclusions ensured that the prototype solved the problem.

The introduction

Egypt faces multiple grand challenges that undeniably slow its development. One of these is the increasing amount of unrecycled e-waste in the environment. As of 2022, the amount of e-waste generated in Egypt is estimated to be ~690,000 tons, which is 17.76% higher than the ~586,000 tons generated in 2019. While reports cite roughly 150,000 collectors + ~50,000 waste processors active in Egypt, the documented formal collection is extremely low in some national studies, as one assessment reported only ~1,584 tons of e-waste is being collected every year, which is only around 2-3% of the yearly generated amount. In addition, only around 20% may undergo local recycling, while the remainder is sent to uncontrolled landfills

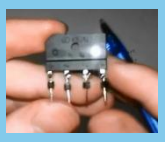
or dumpsites or stockpiled. All these amounts of unrecycled e-waste result in high heavy-metal levels (Pb, Zn, Cu, Mn) in the soil, more specifically in industrial zones. Some studies measured Pb values to reach as high as ~470 mg/kg, and around a major lead smelter site in Egypt, researchers measured extremely elevated Pb in nearby soils (example peak ~3,130 mg/kg).

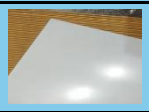
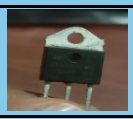
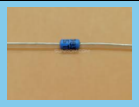
Therefore, our selected problem to be solved is to increase the capabilities of research in Egypt's laboratories through recycling electronic waste. In the meantime, e-waste is one of the severe crises that threatens the residents who live near industrial zones. Hence, solving this problem would successfully reduce the amount of e-waste and heavy metals in the environment, in addition to turning it into a useful project that can be used in school laboratories.

To gather as much beneficial data for the project as possible, different previous solutions for the problem were reviewed. The most outstanding of which were the VEVOR 20L lab incubator and the Thermo Scientific Heratherm IGS180. The common points of strength present in both were the ability to generate temperatures where most types of bacteria can survive and grow ordinarily, and the ability to operate silently.

After reviewing the prior solutions, the selected solution is to construct a lightweight bacterial incubator using unrecycled electronic waste. The incubator will meet a design requirement criterion of performing at least one laboratory application, in addition to having two extra functions.

The materials and methods

Picture							
Name	PTC Aluminum Heater Plate	Electric Transformers 12V & 16 V	Thermoelectric Cooler	RECTIFIER BRIDGE	Home AC unit Thermostat	PC Heatsink and Fan	Old Oven Frame
Quantity	2 pieces	2 pieces (16 V 10 A, and 12 V 1 A)	1 piece	2 pieces	1 piece	1 piece	1 piece

Description	a heating element of the convection system of the prototype	Turns the unstable alternating current found in households into usable DC.	A cooling element of the convection system of the prototype	Turns the Alternating (AC) current into direct current (DC)	To control the temperature of the prototype	Used for heating and cooling in the convection system of the prototype	The frame of the prototype
Picture							
Name	Recycled acrylic	Triac bt136	Capacitors	Potentiometer	Wires	silicon diodes	Diac db3
Quantity	1 Pieces	1 piece	2 pieces (4500 micro 50 volts, and 470 micro 25 volts)	2 pieces	From 1 to 1.5 m	2 pieces	1 piece
Description	To make the side of the frame	Used in the dimmer circuit to control the light intensity	Softens Direct Current (DC) coming out of the rectifier bridges	Two potentiometers one with 5 kilo ohms and other with 1 mega ohm	A copper wire to connect the circuit	Used for setting the speed of the fan for the heating and the cooling system separately	Used in the dimmer circuit to control the light intensity

The methods

1. Disassembled the oven frame, exposed the fiber glass, extracted the fiber glass insulation layer from the oven, and Extracted the PTC heaters and rotary switches of the oven.
2. Removed non-essential components remaining in the oven, cleaned the empty frame of the oven of left over residue **as shown in Figure 1.**
3. Extracted the power supply from the PC to extract the capacitor, wires and the variable resistance.
4. Extracted the rectifier bridge, a capacitor, and the acrylic from the mini fridge.



Figure 27: cleaned the empty frame of the oven

5. The Thermostat was extracted from a home AC unit, and the transformer 16V 10A was extracted from the oven while the 12V 1A transformer was extracted from an old radio.
6. Connected the transformer with the capacitor and the bridge to soften the direct current, put the heat sink in the oven, and the fan.
7. Put the heater plates on the side of heat sink, **as shown in Figure 2**, and the thermoelectric on the back of the oven. Connected the thermostat to the prototype panel and connected silicon diodes on the PTC and thermoelectric cooler wires to the fan to adjust fan speed using variable resistance.
8. Soldered a dimmer circuit from a capacitor, triac, diac, resistor, and variable resistance to control the light intensity, and connected it to an old potentiometer cap from the oven.
9. Established a joint connection between one side of all components, connected the remaining sides to an old rotary switch from the oven.
10. Connected the PTC and thermoelectric wires to the relay of the thermostat.
11. Connected the heat circuit to the transformer output.
12. Completed the final connections of the circuit as shown in the schematic **in figure 3**



Figure 28: Put the heater plates on the side of heat sink

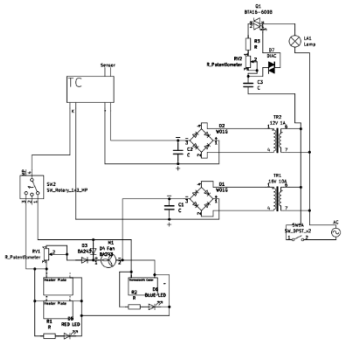


Figure 29: The MultiCubator circuit

Safety precautions

When building the incubator prototype, strict safety precautions have been followed to minimize risks and ensure reliable operation. All electrical components have been properly insulated and grounded to prevent electric shocks or short circuits. Heat sources have been carefully regulated using thermostats and temperature sensors to avoid overheating or fire hazards. The structure has been made from heat-resistant, non-toxic materials to ensure user and sample safety. Proper ventilation has been included to prevent the buildup

of excess heat or harmful gases. Additionally, regular testing, the use of gloves, and other protective equipment during assembly have further improved overall safety.

Test Plan:

A test plan ensuring the solution is both working properly and meeting the design requirements consisted of the following:

1. Using a voltameter, the voltage of the electric current at the point of the PTC heaters and the TEC cooler is measured, ensuring the functionality of the transformer.
2. The temperature in the incubator was measured by putting a thermometer inside it for 5 minutes, ensuring the calibration validity of the heat controller thermostat.
3. A digital thermometer is installed on the top of the incubator's frame, ensuring the quality of the insulation.
4. An E. coli bacteria sample is prepared and placed inside the incubator for 6 hours, testing the growth rate.

Data collection

ITEM	How is it used	Picture
thermometer	To ensure the accuracy of the heat control in the prototype	

Light microscope(x400)	Used in the calculation of growth of the bacteria	

Results:

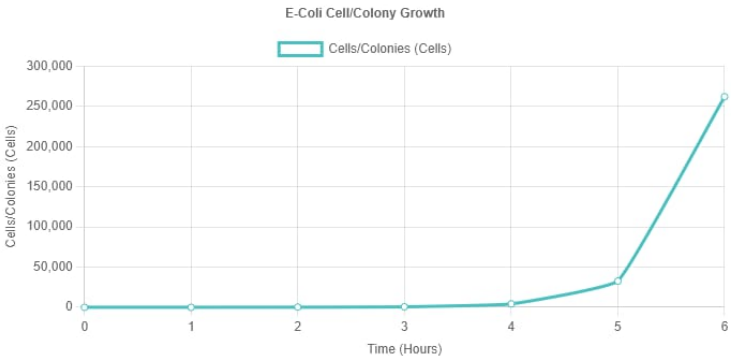
Negative results:

- While disassembling the conventional oven, the frame almost broke due to trying to unscrew it from the wrong side. It was then reassembled, and the legs were properly removed first.
- When the voltage was measured, it was found that the transformer's capacity was not enough for the PTC requirements, so multiple other transistors were used until one worked.
- During the calibration test, it was found that the calibration was off by 0.9°C, then 0.4°C. the thermostat was recalibrated until it matched the real temperature.

Positive results:

- The voltage of the current was exactly 12V, matching the PTC heaters.
- The thermostat calibration matched the thermometer's readings perfectly.
- The digital thermostat measured no change in temperature outside of the incubator, indicating that the insulation is working.
- The measured growth of the E.coli bacteria sample showed exponential growth, which is presented in the following table and graph.

Time (h)	Cells/colony
0	1
1	8±4
2	64±8
3	512±16
4	4,096±32
5	32,768±64
6	262,144±127



Chapter 4

Analysis and Discussion

Binary fission formula

The binary fission formula, mathematically called the exponential growth equation, is the formula used for calculating the number of cells in a colony of bacteria after a certain number of generations or binary fission that happened over an interval of time **as represented by the following figure**. Usually, after just a couple of hours, it becomes impossible to count the cells with just a microscope, so this formula is the main way to calculate the growth of bacteria in low-end laboratories.

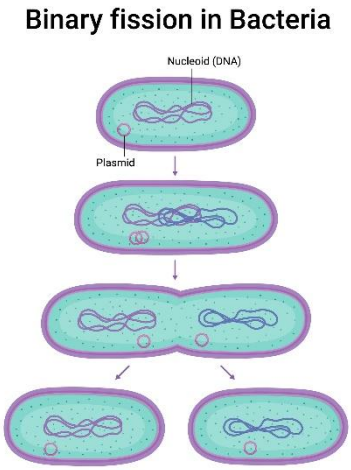


Figure 30: binary fission

$$N_t = N_0 \times (2)^n$$

The N_t is the number of cells after a given amount of time, N_0 is the number of the original cells at time 0, and the n is the number of generations, which is the number of times the cells have undergone the process of binary fission in the studied time period. The number of times the cells divide per hour varies depending on the type of bacteria and the temperature at which they're incubated. To calculate the growth of the bacteria, the n of our bacteria, which is E. coli, should be calculated first. By solving for $n, n = \frac{\log(N_t) - \log(N_0)}{\log(2)}$. Starting with one strain of E. coli bacteria, N_0 is equal to only 1 cell. The sample is then incubated for 1 hour at 37 °C. After incubation, the sample is observed under the microscope with 400x magnification. It is found that there are 8 cells in the sample, meaning that N_t is equal to 8 cells. Substituting in the formula, $n = 3$, meaning the cells have divided 3 times, 1 per 20 minutes. Using the original formula, we can determine how many cells per micro-colony will be present after 6 hours of incubation, which is found to be ~262,144 cells per micro-colony.

Turns ratio equation

The turns ratio equation is the relation between the number of turns revolving around each side of an electrical transformer and the voltage of the current before and after the transformation, **as shown in the figure**. This equation is primarily used to determine the ratio between the turns in a transformer in order to produce the desired Voltage and Amperage for the circuit.

$$\frac{N_p}{N_s} = \frac{V_p}{V_s}$$

In an electrical transformer, the circuit is connected by wrapping the wires around an iron core, creating a magnetic field, where the other side of the wires receives the charge through the magnetic field. The ratio between the number of turns determines how much voltage is produced in the circuit. In the equation, Np is the number of turns in the wire at the side of the power source (the wall outlet), while Ns is the number of turns at the other side. The Vp is the voltage produced by the power source, and Vs is the voltage produced by the transformer. In the incubator, two transformers are needed, one for the thermostat and one for the heater/cooler. The wall outlet produces a voltage of 220V, while the required voltage for the heater/cooler is only 12V. Using the equation, the ratio between the turns needs to be 220/16, which means that for each turn at the heater/cooler's side, 14 turns are needed at the power source side. For the thermostat's transformer, the ratio is 220/12, needing 18 turns at the side of the power source for each one of the other side. Overall, the transformers produce 16V 10A for the heater/cooler, and 12V 1A for the thermostat

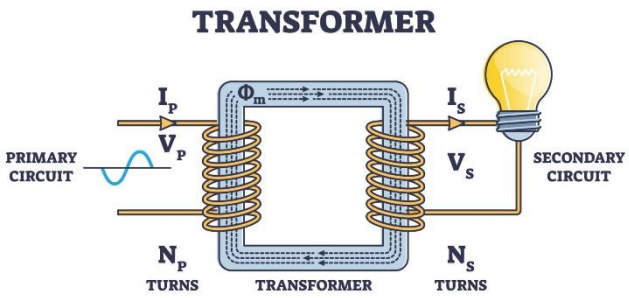


Figure 31: Diagram of a transformer

The Peltier law equation

Standard incubators turn on the heater until the incubator reaches the desired temperature, and when the temperature exceeds the limit, the heater just turns off and waits for the room to lower the temperature. When the room's temperature is higher than the desired temperature, it becomes impossible for the incubator to function, killing the cells. to add a cooling system to the incubator, the Peltier effect is required. The Peltier effect law equation is used to determine the required current intensity in order to reach certain temperatures. The law states that

$$Q = \pi \times I \times t$$

Where Q is the heat energy that is absorbed after the current is applied **as shown in figure 32**, π is the Peltier coefficient, which depends on the material that is used as the Peltier, I is the current intensity passing through the Peltier, which is measured in Ampere, and finally, t is the amount of time the current is passing through the Peltier, where increasing the time would increase the absorbed heat energy simultaneously, since the Peltier continues to absorb the heat energy over time. However, to keep in mind that the electricity passing through the Peltier also creates waste heat, which is calculated by the Joule penalty law: $Q_{waste} = I^2 \times R$, and if Q waste becomes higher than the absorbed heat, Peltier stops cooling. We needed to know the current intensity required to absorb 24,000J, which is equivalent to 15 °C, in only 16 minutes. Solving for I and considering the Joule penalty, it is found that the required current is 10A.

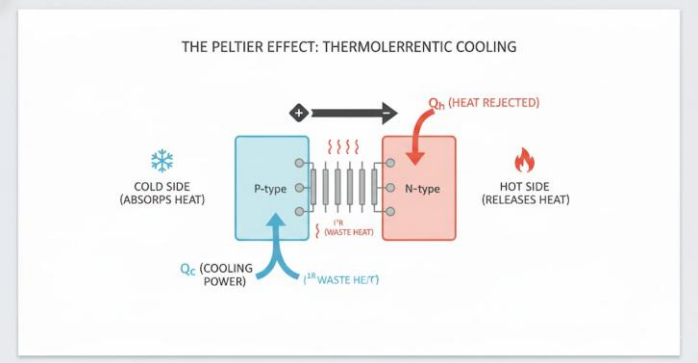


Figure 32: Peltier system

The first law of thermodynamics

The First Law of Thermodynamics is the universe's ultimate accountant. The law of Conservation of Energy applies to thermal systems. The simple statement of the law is that energy cannot be created or destroyed; it can only

change forms. For example, the heat in an incubator is not created, but it is a conversion of electricity into heat. The formula for the law is: $\Delta U = Q - W$, where (ΔU) is the change in internal energy, (Q) is the heat added to the system, and (W) is the work done by the system. The electricity is first converted into heat by the thermostats, then transferred to the air. Since the chamber is a rigid box, the air cannot push through the walls; hence, the work done is equal to zero. This caused the air molecules to move and collide faster, resulting in storing the heat in the air and transferring it to the bacterial sample. If the heater runs for 10 minutes, let the power generation rate be 160 J/s, and the time be 600 seconds, meaning that the energy added to the system is equal to 96,000 J throughout the 10 minutes. and the change in internal energy, which is calculated using the Peltier effect, is equal to 24,000J. According to the first law of thermodynamics, the work done, which is the leakage of energy in the system is = 96,000 - 24,000 = 72,000J. Since the time is 600s, this means that the power leakage is about 120J/s.

The table of learning outcomes

Subject	Learning outcome	Key concept	Usage
Maths	Math.2.03	The Permutations	To calculate the uncertainty in the growth of bacteria
Maths	Math.2.01	Polynomial functions	To draw the data and expect the results and the range of a specific data set
biology	BIO.2.01	The GMO	Helped us know how the genetic material of different organisms affects their preferred habitats
Physics	PH.2.03	Dynamic electricity	To know the required conditions for the continuous flow of electric charge in the circuit.
Physics	PH.2.05	Capacitor charging and discharging	To study the capacitors to use them to soften the circuit

The conclusion

The problem chosen to be solved for this project was to turn electronic waste into a beneficial device able to serve in school laboratories. The selected solution was to construct a bacteriological incubator, as it has the potential to utilize multiple different wasted electronic parts and make a real difference in the school's biology lab. The device was made of a heating system, a cooling system, and an old oven functioning as the shell of the prototype. Regardless of the negative results, the prototype has successfully met all the major design requirements of performing at least two functions (generating heat and controlling light intensity) and providing at least one measurement (the internal chamber's temperature). Moreover, the device was able to overcome the high initial price problem present in the two previous solutions, the VEVOR 20L lab incubator and the Thermo Scientific Heratherm IGS180. By reviewing and benefiting from other solutions, the device has successfully increased the research capabilities in school labs, besides turning e-waste into something beneficial.

Recommendations

1. Humidity control system recommendation

It is highly recommended to add a humidity control system to the device due to its ability to measure humidity levels in the chamber and make adjustments as needed. Modifying the incubator to utilize this system would enable it to preserve the agar from evaporating for a longer duration, allowing for longer research periods without needing to replace the agar plates frequently. However, implementing this suggestion is challenging due to safety concerns and material scarcity. Humidity sensors are extremely rare to find in older devices, and even if they can be acquired, they pose potential safety risks. Humidity control systems require a strong sealing mechanism to minimize moisture exposure to near heating elements and electronics. Otherwise, water could damage the circuitry and potentially pose a hazard to users. A final concern to address is accuracy. Constructing a humidity system using scavenged materials while maintaining an error percentage within ±5% would be an extremely challenging task.

2. CO2 injection system:

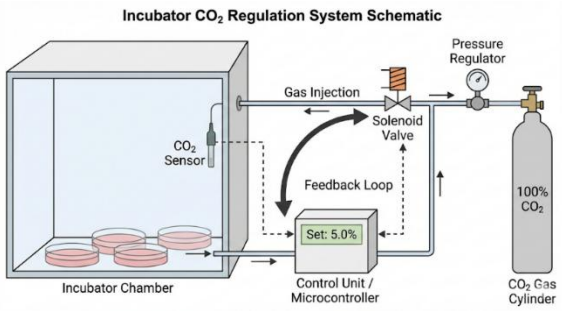


Figure 33: CO2 injection system

An important addition to be recommended for the incubator is a CO2 injection system. This is because most of the culture media contain sodium bicarbonate, which is a base. The CO2 reacts with the water, which forms carbonic acid. This neutralizes the media. Otherwise, the system would lose acidity, and the cells would eventually die in the long term. The CO2 injection system consists of an IR sensor that detects the CO2 levels in the incubator, and when it drops below 5%, it opens a solenoid valve that's connected to a CO2 gas cylinder, puffing the gas into the incubator. **This system is presented in the following Figure(33).** We unfortunately couldn't apply such a system in our prototype, since, without a license, we are not allowed to have a CO2 container for safety reasons.

3. Water jacket insulation system:

The most important section in the incubator is the insulation system, since poor insulation would lead to huge amounts of heat losses, which would put the heaters under a higher load just to reach the desired temperature. For instance, if the heaters worked regularly, they would need to be active every 3 minutes; poor insulation would drastically increase that to once every minute. This also has a higher risk of killing the cells, since if the walls get too cold, water droplets will start to build up on the walls and the ceiling, and if a droplet falls on the petri dish, this instantly contaminates and kills the cells. We can use the water jacket insulation system to insulate the incubator, which works by double-layering the walls and pouring warm water in between. Water has high heat capacity, which is why it's a good insulator, and it also matches our requirement of the project being environmentally friendly. We instead used passive foam, since the water jacket requires stainless steel, which is hard to find in old devices.

4. stainless steel & copper lining:

it's recommended to double-layer the body of the incubator, where it's best to use stainless steel for the outer shell, which has high chemical resistance, preventing rusting. However, for the inner chamber and racks, it's recommended to use copper lining as the main material, **as shown in the figure 34.** Biofilm has a high risk of cross-contamination between the bacteria, where the bacteria produce an extracellular matrix that sticks to surfaces, and rough surfaces like wood or plastic are easier to stick to. To prevent biofilm, the best way is to use copper lining. This totally prevents any leakage of bacteria outside the petri dish. That's because copper lining is oligodynamic, which means that it instantly kills any bacteria that touch it. This is crucial, since the one thing that shouldn't happen in a laboratory is contamination in/outside the incubators, which can cause catastrophic disease spreads. This wasn't applied in the prototype because copper lining costs too much, reaching 11 USD per kg.

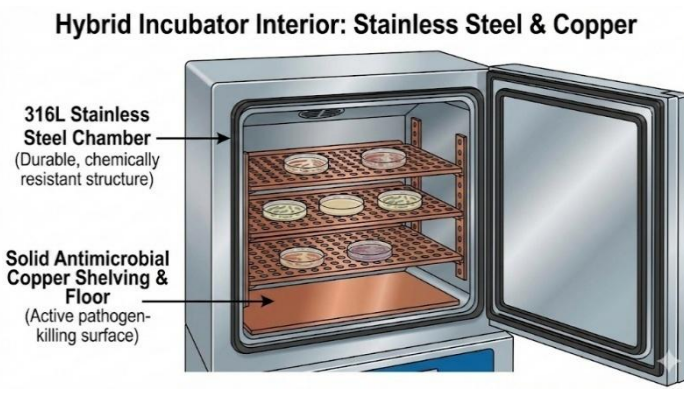


Figure 34: Copper lining

Recommendations for future teams

We, team 209, have gone through many difficulties in the process of finalizing this project and solving as many challenges as can be covered by our idea, the biological incubator. With that in mind, we have found it to our greatest concern to guide the upcoming seniors who might have a similar project idea to our project to elevate the efficiency of their research.

The most notable difficulty that has hindered our progress was the lack of materials necessary to conduct the test plan of our prototype. After extensive research, we reached out to a doctor in an Egyptian university who kindly agreed to help us acquire the unavailable materials. We recommend that future teams who face this issue immediately contact universities and/or research centers as we did. As many materials needed, such as E. coli bacteria, are nowhere to be found in the market.

In addition to the latter, we recommend future teams constructing a biological incubator or a similar prototype to extend their research on the efficiency of cooling elements if any will be used, especially if the chosen element is of thermoelectric nature as they, from our experience, perform less than expected.

Project Benefits

The MultiCubator, at its core, is a biological incubator meant to be used in high school biology laboratories for use in academic research in multiple fields such as genetics and microbiology. However, it is not limited to that function. The MultiCubator can also function as a dough fermenter, egg incubator, and simple cooler. Consequently, the benefits of this project vary from laboratory usage to normal day-to-day tasks.

The first benefit of this project would be its versatility as it can be used to preserve bacterial samples or other chemicals or substances at certain temperatures at school laboratories. The project can also be used at homes to boost the fermentation process of dough while baking or preserving food. Not only that but it can also be used at chicken farms to incubate eggs, with slight modifications.

Another benefit of the MultiCubator is its simple design and user-friendly control system. The prototype is constructed with a straightforward design with minimal components and a self-explanatory control system which allows it to be used by high school students without any prior experience with biological incubators. In addition to that, simple controls also limit the risks of misuse, which in turn decreases hazards and/or sample contamination.

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